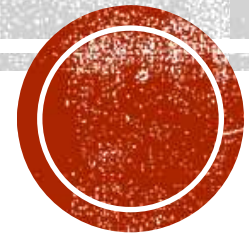


INTERRUPTED TIME SERIES ANALYSIS IN ENVIRONMENTAL EPIDEMIOLOGY: AN OVERVIEW OF TRADITIONAL AND NOVEL MODELING APPROACHES

30 August 2026

ISEE pre-conference workshop

Yiqun Ma and Tarik Benmarhnia



AGENDA FOR TODAY

- Logistics and introductions
- Part I Tarik ~ 80 min
- Q&A
- Break (~5-10min)
- Part II Yiqun ~ 80 min
- Q&A
- End



WORKSHOP OBJECTIVES

1. Understand the conceptual foundations of quasi-experimental methods based on the timing of a natural experiment
2. Understanding how Interrupted Time Series (ITS) and extensions work with a focus on specific identification assumptions.
3. Being able to implement (in *R*) ITS in different settings, interpret estimands of interest and check/visualize possible assumptions' violations → Part II



OUTLINE

- Causal inference, natural experiments and quasi-experimental methods
- Difference-in-differences methods
- Interrupted Time Series
 - Traditional 1-stage approaches
 - 2-stage ITS
 - The *its2s* package
- Focus on staggered interventions
 - Case study: understanding heterogeneity of wildfires impacts in California
 - Tropical cyclones in Taiwan



WHAT DO EPIDEMIOLOGISTS DO?

- **Describe** patterns of injuries, disease, and other health outcomes
- **Predict/suggest** [clinical, individual, or population] different patterns over space/time
- **Understand the causes** that produce those patterns [etiology]
 - **Evaluate** the effectiveness of interventions [etiology too]



RANDOMIZATION AS A SOLUTION TO DEAL WITH CONFOUNDING

- Randomization has been proposed as a solution to deal with confounding
 - Complying with exchangeability between exposed and non-exposed to the policy/treatment
 - For both measured & unmeasured confounding



IDEAL RANDOMIZED EXPERIMENTS

- What makes an ideal randomized experiment:
 - No loss to follow-up,
 - Full adherence to the assigned treatment over the duration of the study,
 - A single version of treatment, and double-blind assignment
- Ideal randomized experiments are unrealistic **but useful to introduce some key concepts for causal inference**



THE IDEA BEHIND RANDOMIZATION

- How do RCT work ...
- Randomization can be analyzed as an “**Instrumental Variable**”
- The overall aim is to deal with both measured and unmeasured confounding



- **The main problem:** Most of the time RCT are not feasible



WHY RCT ARE NOT SUFFICIENT?

- The parachute use example
- RCTs are sometimes impossible for:
 - Economic reasons
 - Logistical reasons
 - Ethical reasons
- Sometimes it is simply too late
- Other misconceptions about RCTs



Understanding and misunderstanding randomized controlled trials

Angus Deaton^{a,b,c,*}, Nancy Cartwright^{d,e}

BMJ VOLUME 327 20-27 DECEMBER 2003 bmj.com

Hazardous journeys

Parachute use to prevent death and major trauma related to gravitational challenge: systematic review of randomised controlled trials

Gordon C S Smith, Jill P Pell



Parachutes reduce the risk of injury after gravitational challenge, but their effectiveness has not been proved with randomised controlled trials

Abstract

Objectives To determine whether parachutes are effective in preventing major trauma related to gravitational challenge.

Design Systematic review of randomised controlled trials.

Data sources: Medline, Web of Science, Embase, and the Cochrane Library databases; appropriate internet sites and citation lists.

Study selection: Studies showing the effects of using a parachute during free fall.

Main outcome measure Death or major trauma, defined as an injury severity score > 15 .

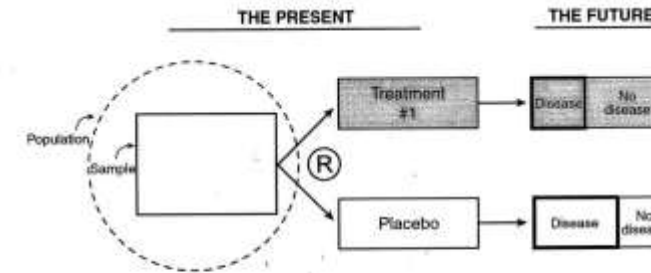
Results We were unable to identify any randomised controlled trials of parachute intervention.

Conclusions As with many interventions intended to prevent ill health, the effectiveness of parachutes has not been subjected to rigorous evaluation by using randomised controlled trials. Advocates of evidence based medicine have criticised the adoption of interventions evaluated by using only observational data. We think that everyone might benefit if the most radical protagonists of evidence based medicine organised and participated in a double blind, randomised, placebo controlled, crossover trial of the parachute.



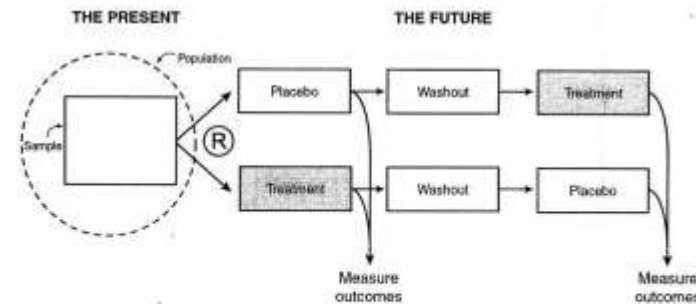
DIFFERENT TYPES OF RCTS

- Two-arm, parallel design
- Planned cross-over design
- Factorial design
- Cluster randomized trials
- **Stepped Wedge Designs**



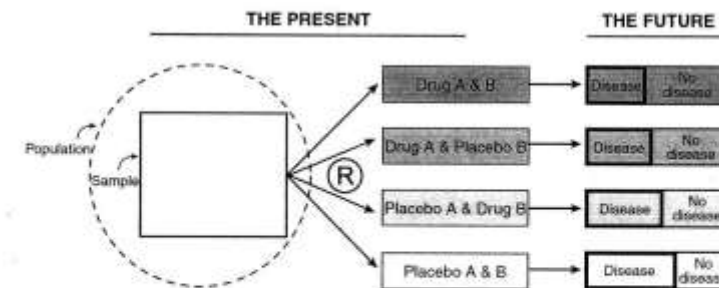
Two-arm, parallel design

■ **FIGURE 10.1**
In a randomized trial, the investigator (a) selects a sample from the population, (b) measures baseline variables, (c) randomizes the participants, (d) applies interventions (one should be a blinded placebo, if possible), (e) follows up the cohort, (f) measures outcome variables (blindly, if possible) and analyzes the results.



Planned cross-over design

■ **FIGURE 11.4**
In the cross-over randomized trial, the investigator (a) selects a sample from the population, (b) measures baseline variables, (c) randomizes the participants, (d) applies interventions, (e) measures outcome variables, (f) allows washout period to reduce carryover effect, (g) applies intervention to former placebo group, (h) measures outcome variables again.



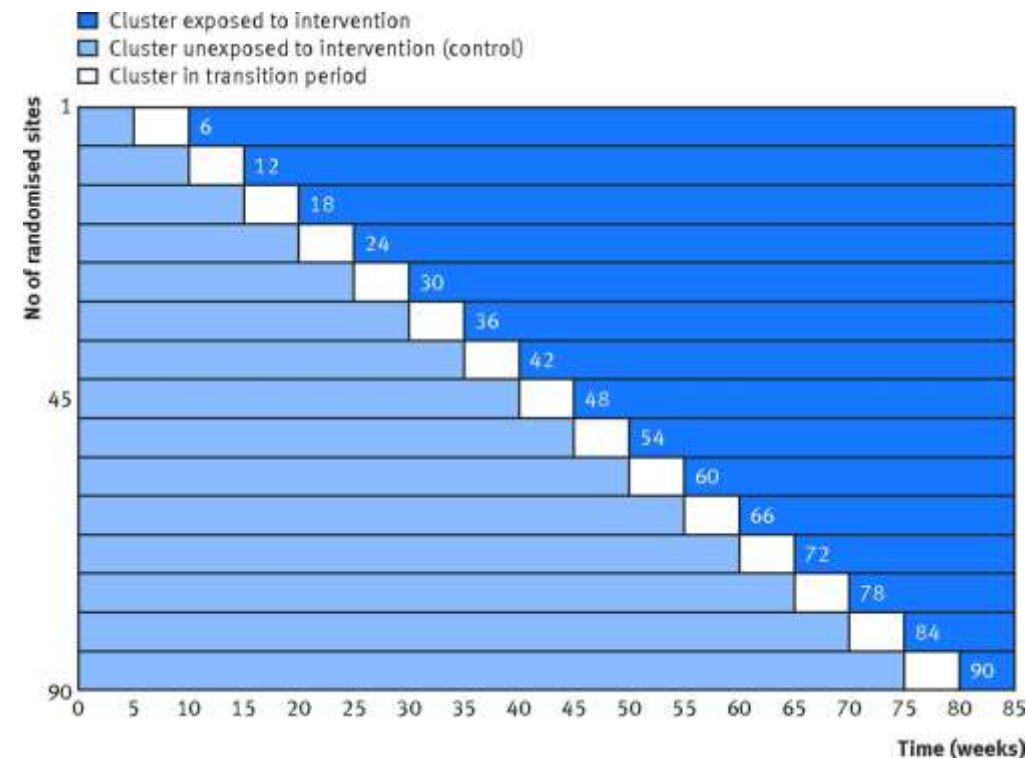
Factorial design

■ **FIGURE 11.2**
In a factorial randomized trial, the investigator (a) selects a sample from the population; (b) measures baseline variables; (c) randomly assigns two active interventions and their controls to four groups, as shown; (d) applies interventions; (e) follows up the cohorts; (f) measures outcome variables.



DIFFERENT TYPES OF RCTS

- Two-arm, parallel design
- Planned cross-over design
- Factorial design
- Cluster randomized trials
- **Stepped Wedge Designs**
 - **Based on the timing of the intervention**



Hemming, Karla, et al. "The stepped wedge cluster randomised trial: rationale, design, analysis, and reporting." *Bmj* 350 (2015): h391.



THE TARGET TRIAL FRAMEWORK

- Target trial emulation is the application of design principles from randomized trials to the analysis of observational data, thereby explicitly tying the analysis to the trial it is emulating.
- The purpose is to improve the quality of observational epidemiology through the application of trial design principles, even when, or perhaps especially when, a comparator trial is not yet available or feasible

Eur J Epidemiol (2017) 32:473–475
DOI 10.1007/s10654-017-0293-4

COMMENTARY

Target trial emulation: teaching epidemiology and beyond

Jeremy A. Labrecque¹ · Sonja A. Swanson¹

Practice of Epidemiology

Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available

Miguel A. Hernán* and James M. Robins

USING NATURAL EXPERIMENTS

- By capitalizing on natural experiments, Quasi-experimental methods (QEM) can be used as alternatives to experimental methods to provide causal estimates from observational studies.
- The term quasi-experiment refers to:
 - *“experiments that have treatments, outcome measures, and experimental units, but do not use random assignment to create the comparisons from which treatment-caused change is inferred”* (Cook et al. 1979)
- Different QEM, different configurations, different assumptions to draw causal inference
- To partially deal with both measured and unmeasured confounding



EXAMPLES OF NATURAL EXPERIMENTS

NOT ONLY FOR POLICY EVALUATION

- Public policies:
 - Smoking ban
 - Legalization of marijuana
- Clinical Treatments, Vaccination
- Conditional Cash Transfers
- **Climate Hazards**
 - Wildfires
 - Hurricanes

2 main types of natural experiments

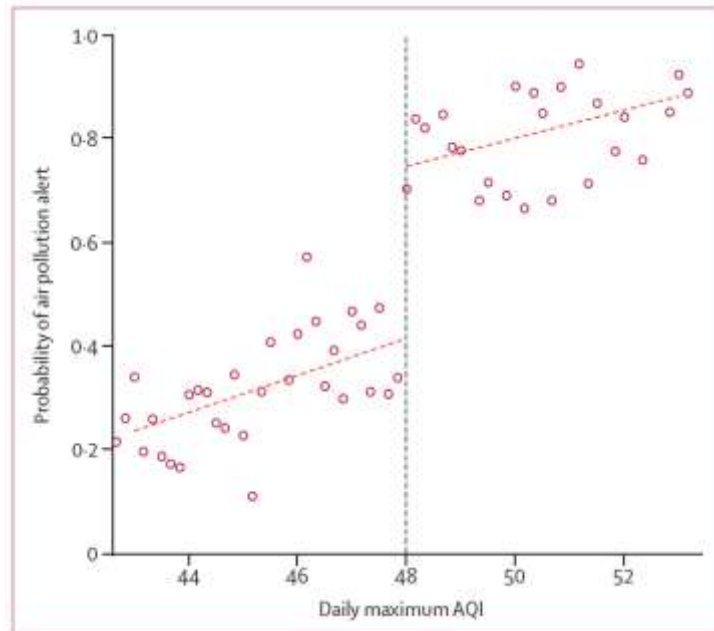
- **Timing of the intervention**
- Eligibility to a specific policy (e.g. age for vaccination) and other IV strategies



USING NATURAL EXPERIMENTS TO EVALUATE THE BENEFITS OF AIR QUALITY INTERVENTIONS

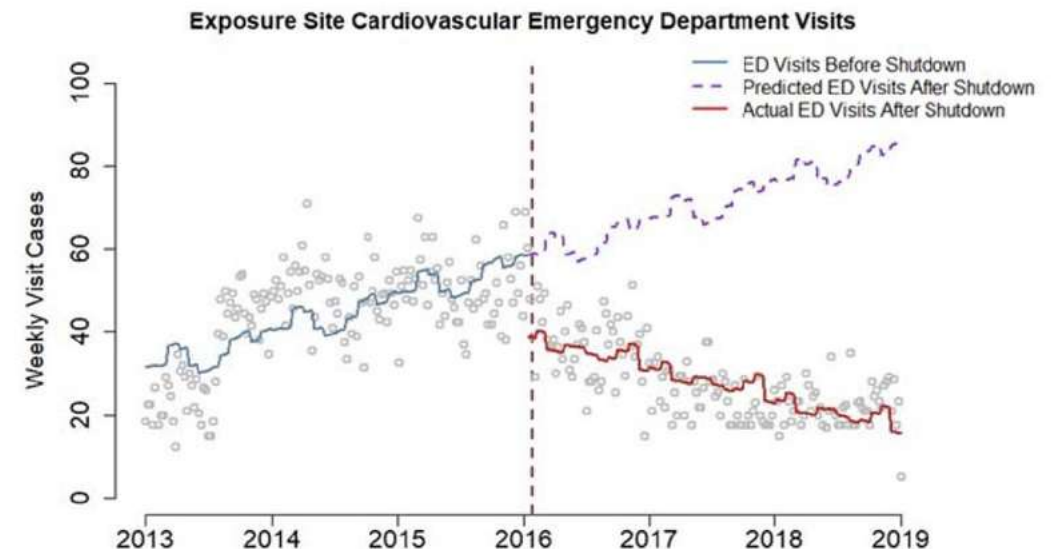
Effect of air quality alerts on human health: a regression discontinuity analysis in Toronto, Canada

Hong Chen, Qiongsi Li, Jay S Kaufman, Jun Wang, Ray Copes, Yushan Su, Tarik Benmarhnia



An interrupted time series analysis of the cardiovascular health benefits of a coal coking operation closure

Wuyue Yu^{*} and George D Thurston^{*}



REGRESSION DISCONTINUITY WITH TIME AS THE FORCING VARIABLE

- Several studies used discontinuity in time as their identification strategy
- Example: Protective Effects of Smoke-free Legislation on Birth Outcomes in England

Protective Effects of Smoke-free Legislation on Birth Outcomes in England

A Regression Discontinuity Design

*Ioannis Bakolis,^{a,b,c} Ruth Kelly,^{d,e} Daniela Fecht,^a Nicky Best,^d Christopher Millett,^f Kevin Garwood,^a
Paul Elliott,^{a,g} Anna L. Hansell,^{a,g} and Susan Hodgson^h*

- In this setting, a regression discontinuity design is comparable to interrupted time series (ITS) analyses
 - Nianogo R & Benmarhnia T. Instrument-Based Methods for Causal Inference in Epidemiologic Research: A Unifying Framework



Regression Discontinuity in Time: Considerations for Empirical Applications

Catherine Hausman¹ and David S. Rapson²

- In their review, Hausman and Rapson, discuss several features of the regression discontinuity in time framework
- The modelling approach in a typical RD analysis and a 1-stage ITS are the same



CONSIDERING CLIMATE HAZARDS AS NATURAL EXPERIMENTS

Using the Generalized Synthetic Control Method to
Estimate the Impact of Extreme Weather Events on
Population Health

Paige Sheridan,^{a,b} Sara McElroy,^{a,b} Joan Casey,^c and Tarik Benmarhnia^{a,d}

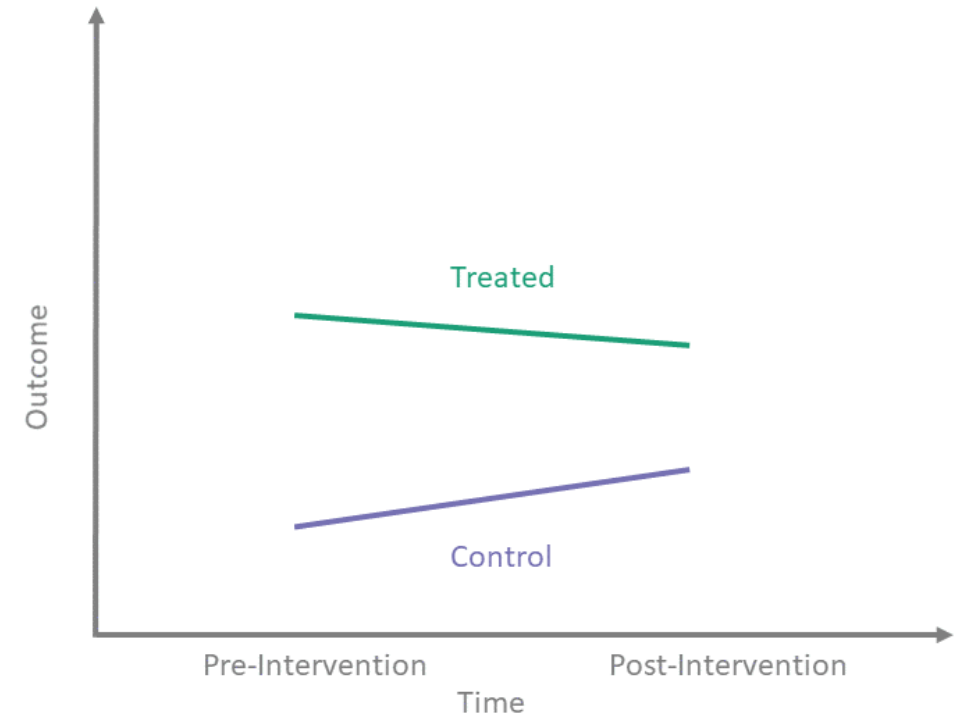
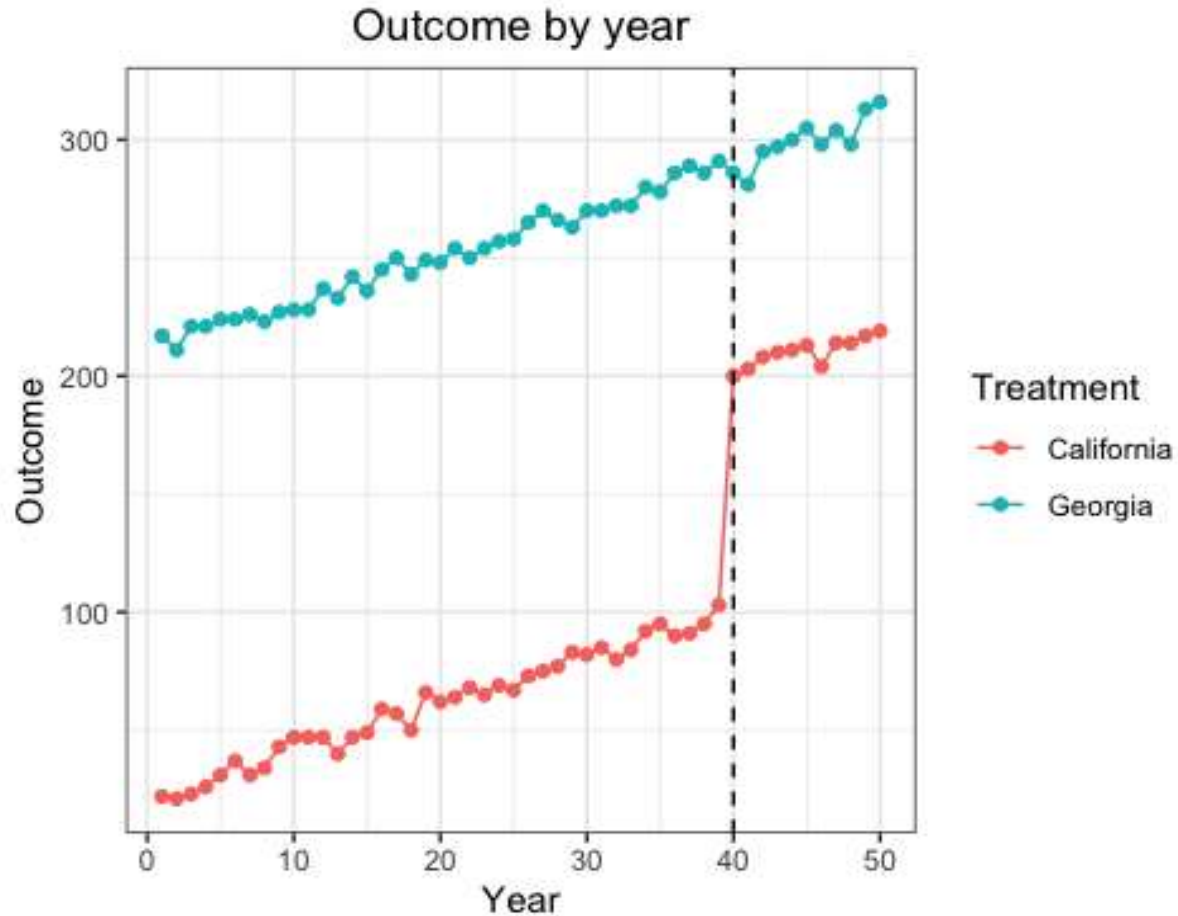
- If we can argue that the timing of a given extreme weather event is independent with the trend for the outcome of interest, we can capitalize on this as a natural experiment
- Multiple approaches exist to capitalize on such natural experiments
 - Difference-in-differences methods
 - Synthetic control methods
 - **Interrupted time series analyses**



DIFFERENCE-IN-DIFFERENCES



The difference-in-differences idea



By Bret Zeldow and Laura Hatfield



DIFFERENCE-IN-DIFFERENCES

THE STANDARD APPROACH

- To estimate the effect of interest

$$(\mu_{11} - \mu_{10}) - (\mu_{01} - \mu_{00})$$

$i = 0$ is control group, $i = 1$ is treatment.

$t = 0$ is pre-period, $t = 1$ is post-period.

- Simple Regression modeling approach (with 2 groups)

- $E[Y \mid a, t, did] = \beta_0 + \beta_1 a + \beta_2 t + \beta_3 did$

- Where:

- a represents the group with the policy
 - t represents the period after the policy implementation
 - DID is the interaction between a and t



HOW DOES IT WORK?

	T=0	T=1
A=0	β_0	$\beta_0 + \beta_1$
A=1	$B'_0 + \beta_2$	$B'_0 + \beta'_1 + \beta'_2 + \beta_3$

$$E[Y \mid a, t, \text{did}] = \beta_0 + \beta_1 t + \beta_2 a + \beta_3 [a * t]$$

0	0
1	0
0	1
1	1

Like a two-way fixed effect (TWFE)

You can also include more complex time trends



DID ASSUMPTIONS

- The key assumptions of the DiD analysis are:
 1. The trend in the control group represents a good approximation for the counterfactual trend of the treated group in the absence of the treatment.
 2. Common Shock Assumption
 3. No spillover



MANY APPLICATIONS

Removing user fees for facility-based delivery services: a difference-in-differences evaluation from ten sub-Saharan African countries

Britt McKinnon,^{1*} Sam Harper,¹ Jay S Kaufman¹ and Yves Bergevin²

Gotta catch'em all! Pokémon GO and physical activity among young adults: difference in differences study

Katherine B Howe,^{1,2} Christian Suharlim,³ Peter Ueda,^{4,5} Daniel Howe, Ichiro Kawachi,² Eric B Rimm^{1,6,7}

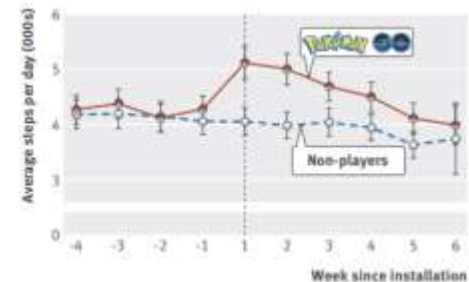


Fig 1 | Average number of daily steps and 95% confidence intervals by week before and after installation of Pokémon GO (median 8 July 2016)

JAMA Pediatrics | [Original Investigation](#)

Difference-in-Differences Analysis of the Association Between State Same-Sex Marriage Policies and Adolescent Suicide Attempts

Texting Bans and Fatal Accidents on Roadways:
Do They Work? Or Do Drivers Just React to
Announcements of Bans?[†]

By RAHI ABOUK AND SCOTT ADAMS[#]

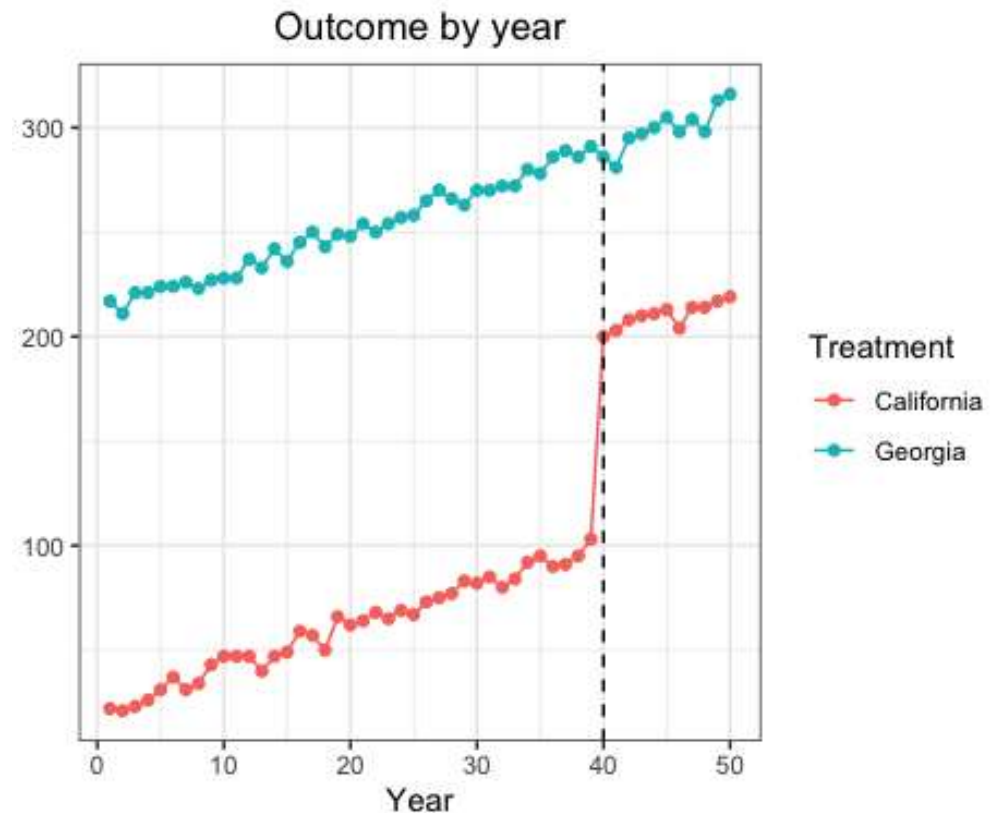
Evaluating the effect of hierarchical medical system on health seeking behavior: A difference-in-differences analysis in China

Zhongliang Zhou^a, Yaxin Zhao^{b,*}, Chi Shen^a, Sha Lai^a, Rashed Nawaz^a, Jianmin Gao^a



WHY DO WE NEED A CONTROL GROUP?

- When capitalizing on the timing of a natural experiment, relying on control groups **helps with the common shock assumption**



CONTROLLED ITS OR DID?

- When using one (or more) control group.s, there is no fundamental distinction between Controlled ITS and DID models
- They yield the same counterfactuals and identify the same treatment effects.
- The terminology simply varies across disciplines



ALTERNATIVE OPTIONS FOR SELECTING CONTROL GROUPS

- Besides geographical units that did not receive the treatment/policy, it is possible to consider different types of control groups
- It is possible to use different outcomes or population subgroups to contrast the change in the outcome over time
 - For example, if a given policy only targets individuals above 65 years, it will be possible to use the 64 years of age and below subgroup as a reference group

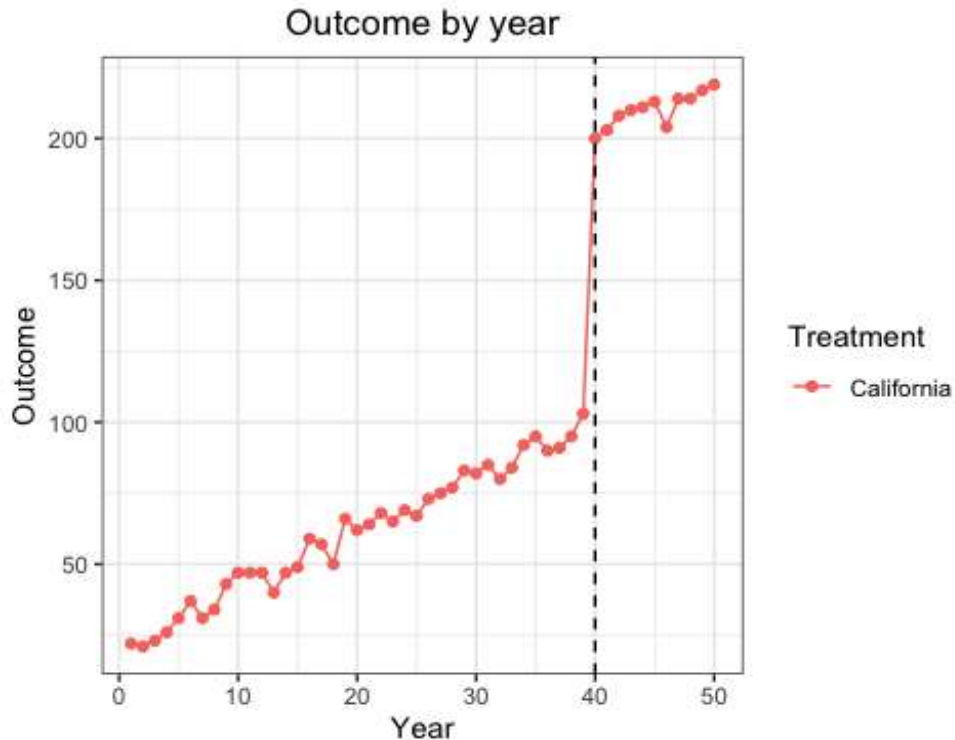


WHEN CONTROL GROUPS ARE NOT AVAILABLE

- Often, it is impossible to obtain control groups because:
 - The entire area has been exposed or received the policy
 - Data is simply not available
- But, by making an additional assumption about the common shock assumption, it is possible to use **interrupted time series** models



INTERRUPTED TIME SERIES (ITS): DID WITHOUT CONTROL GROUPS



Estimation

The traditional 1-stage ITS approach:

The following model can be used to estimate the effect of the policy on the outcome y

$$y = \alpha + \beta_1(\text{year}) + \beta_2(\text{post}) + \beta_3(\text{xi}) + \beta_4(\text{xt}) + \beta_5(\text{year} \times \text{post}) + \epsilon$$

- β_2 corresponds to the differential intercept
- β_5 corresponds to the differential slope

AKA segmented regression models

We model time trends (xt) and time-varying covariates (xi) using traditional time series methods

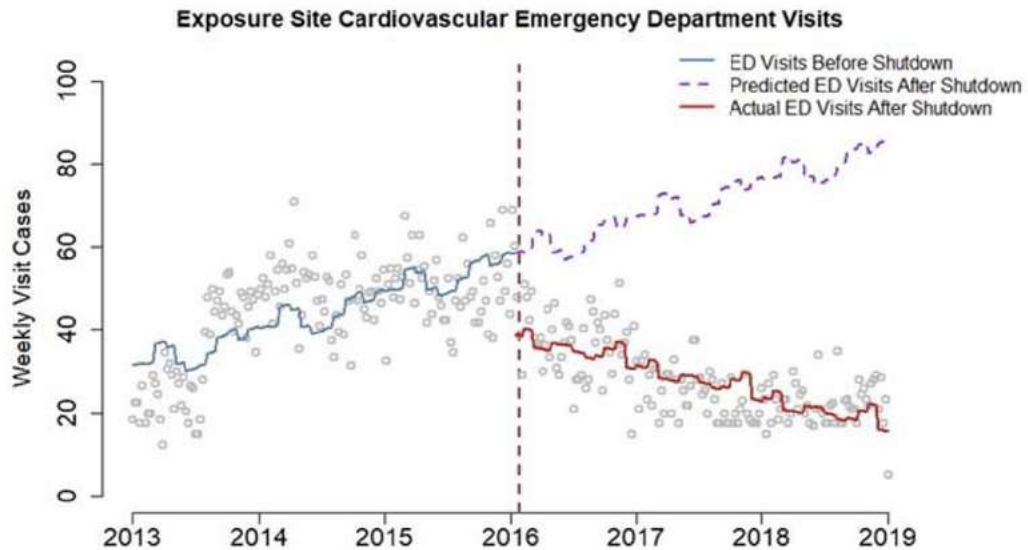
The traditional approach to optimize the model is based on the Akaike Information Criterion (AIC)



1-STAGE ITS EXAMPLES

An interrupted time series analysis of the cardiovascular health benefits of a coal coking operation closure

Wuyue Yu^{*}  and George D Thurston[†] 



INTRODUCING 2-STAGE ITS



TWO-STAGE TIME SERIES ANALYSIS COUPLED WITH MACHINE LEARNING: EVALUATING THE HEALTH EFFECTS OF THE 2018 WILDFIRE SMOKE EVENT IN SAN FRANCISCO COUNTY AS A CASE STUDY

- We compared multiple algorithms in a 2-stage ITS approach
 - Autoregressive Integrated Moving Average (ARIMA)
 - NNETAR (Neural Network)
 - Prophet-XGBoost

Table 1. Performance metrics of the ARIMA, NNETAR, and Prophet-XGBoost model

	Training (2009-01-01 to 2016-11-07)			Testing (2016-11-08 to 2018-11-07)		
	ARIMA	NNETAR	Prophet-XGBoost	ARIMA	NNETAR	Prophet-XGBoost
R ²	0.71	0.83	0.83	0.65	0.58	0.71
MAE	7.08	5.52	5.33	8.67	9.36	8.11
RMSE	9.19	7.12	7.02	11.84	13.00	10.85
MAPE	0.14	0.11	0.11	0.16	0.16	0.15
SMAPE	0.13	0.11	0.10	0.16	0.17	0.15

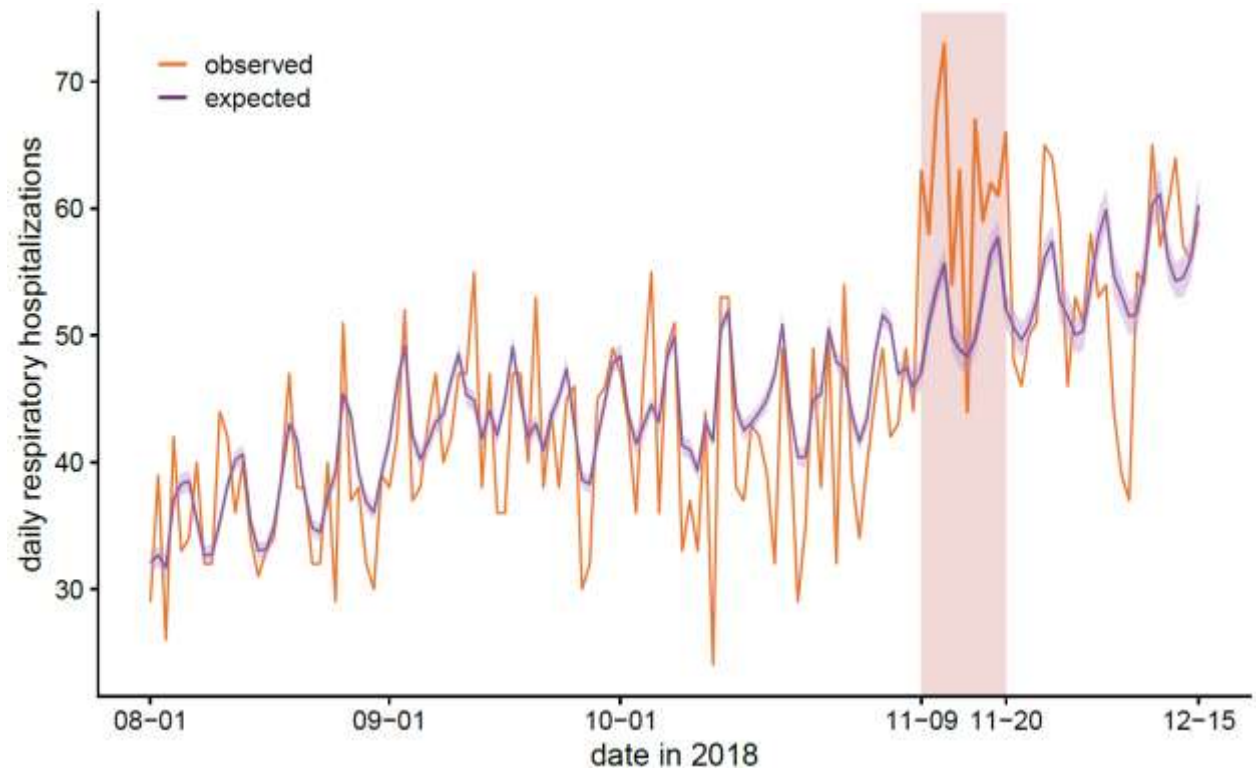
R²: coefficient of determination

MAE: mean absolute error

RMSE: root mean square error

MAPE: mean absolute percentage error

SMAPE: symmetric mean absolute percentage error



WHAT IS MACHINE LEARNING?

- ML algorithms perform quite well for predictive problems
 - Definition: ML Algorithms aim to 'learn' or predict outputs (exposed/unexposed, treated/untreated) from inputs (covariates) in a new sample, having been first trained on a training dataset that contains both inputs and labelled outputs.
 - Blakely, Tony, et al. "Reflection on modern methods: when worlds collide—prediction, machine learning and causal inference." International journal of epidemiology (2019).
- **Cross-validation**, splitting the sample into training and validation (testing) subsamples
- Different types of ML algorithms
 - Random Forest
 - **Gradient boosting**
 - Lasso, Ridge and Elastic net
 - Ensemble Learning (or super learner). : A technique using multiple algorithms (and could include traditional regression methods) that combines them to improve estimates and predictive performance. Include bagging, boosting and stacking



GOOD PRACTICES FOR PREDICTIVE ANALYTICS (NAIMI 2019)

- Principle 1: Identify the Purpose of a Given Prediction Algorithm
- Principle 2: Identify a –realistic- Context of a Given Prediction Algorithm
- Principle 3: Consider the Range of Analytic Methods Available
- Principle 4: Honest Training and Validation

More details during the applied part

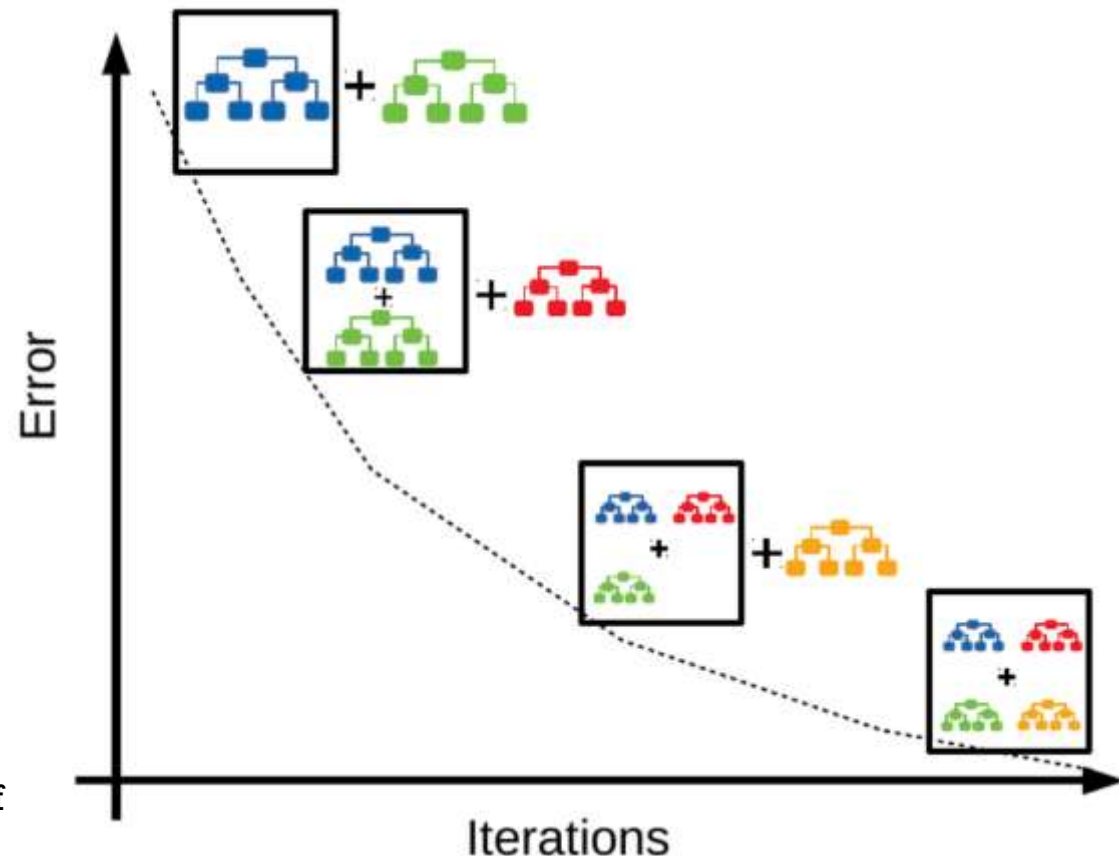


UNDERSTANDING THE INTUITION BEHIND GRADIENT BOOSTING

- Consider the following predictive problem:
 - We want to predict PM2.5 in a given day in a given census tract
 - We have multiple predictors:
 - Temperature
 - Precipitation
 - Green space coverage
 - Traffic flow
 - Calendar variables

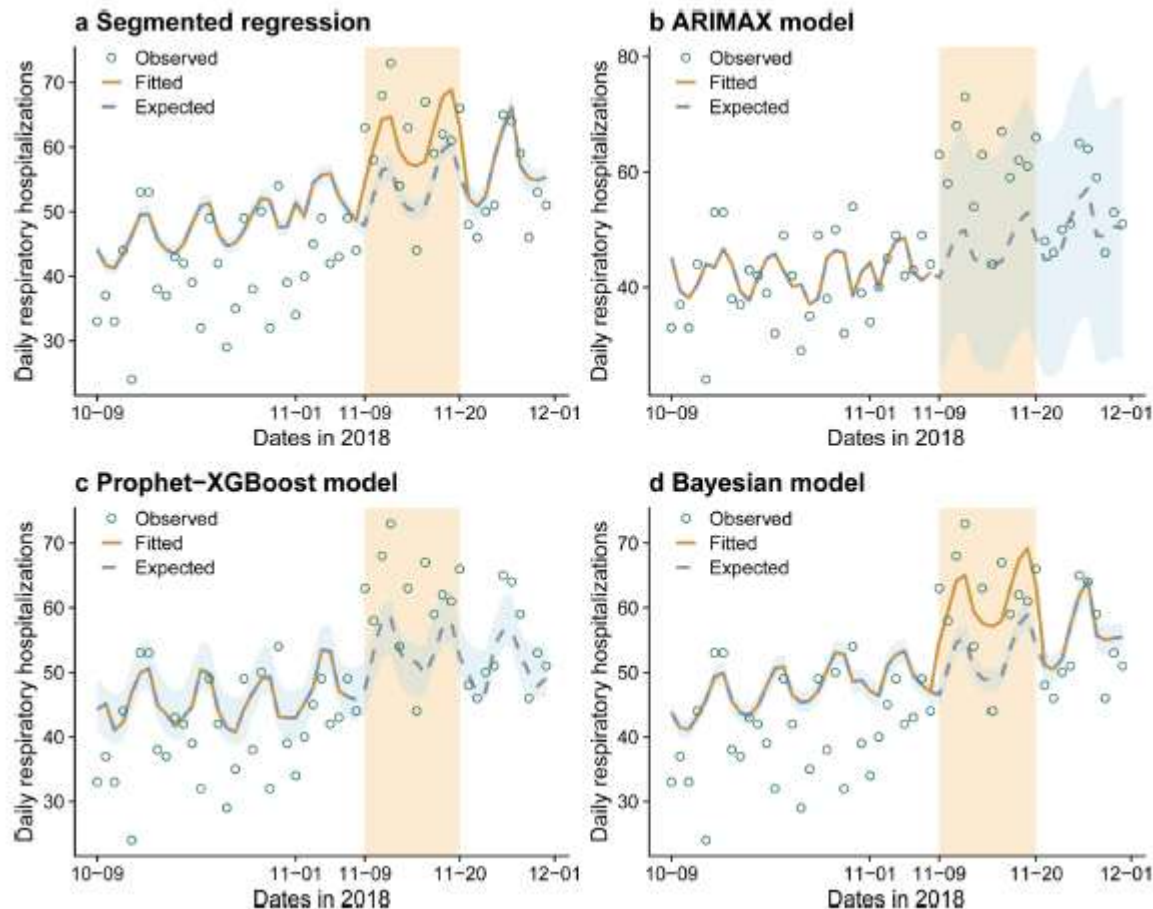
$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

The Root Mean Square Error (RMSE) formula is the square root of the average of the squared differences between predicted and actual value



Interrupted Time Series Analysis in Environmental Epidemiology: A Review of Traditional and Novel Modeling Approaches

Yiqun Ma¹ · Tarik Benmarhnia^{1,2}

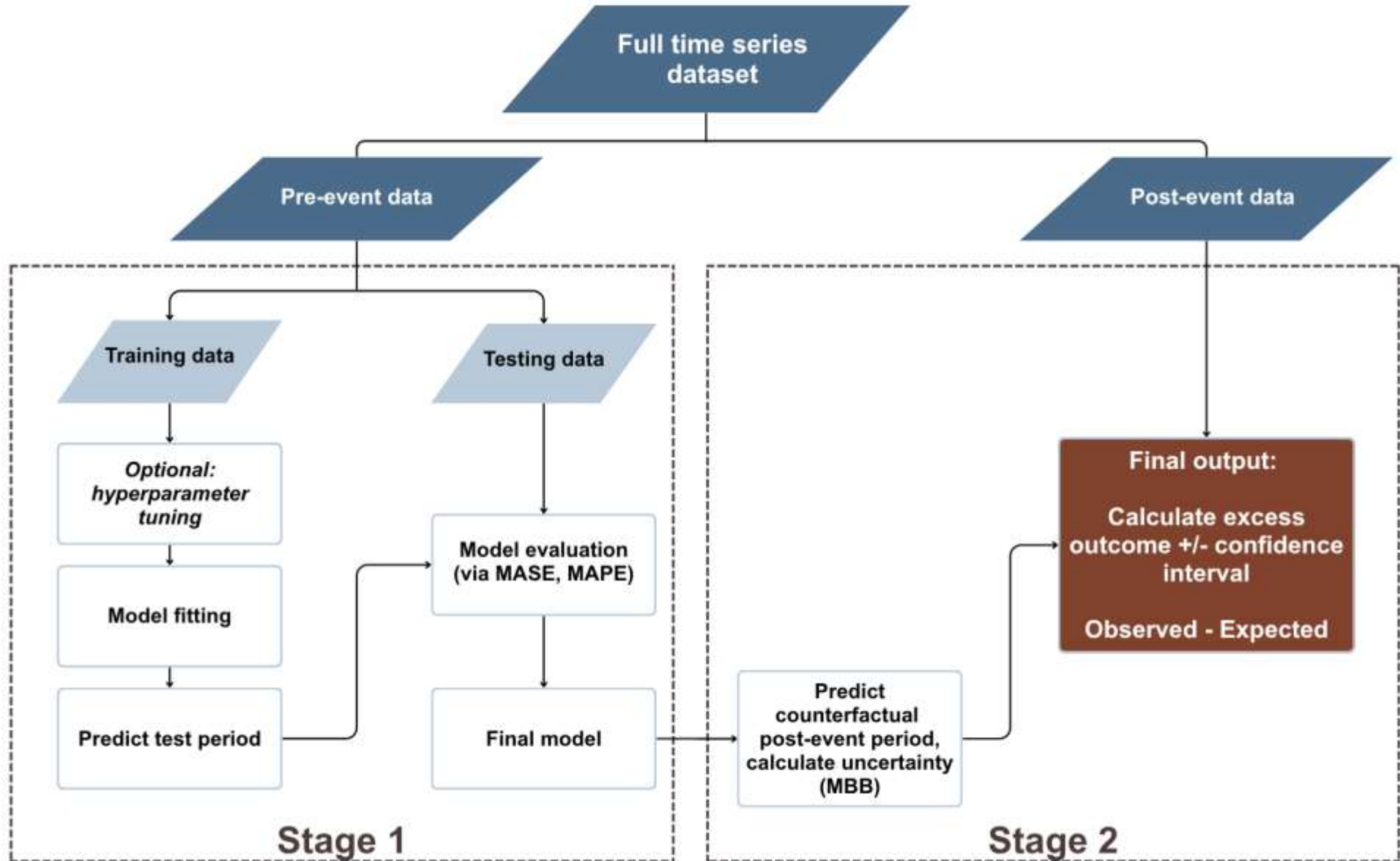


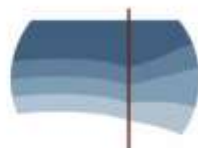
INTRODUCING THE IT2S FRAMEWORK

- Website: <https://itslab.io/>
- Paper: DOI: [10.64898/2026.07.02.26357175](https://doi.org/10.64898/2026.07.02.26357175)
 - Lauren B. Wilner, Joan A. Casey, Stephen J. Mooney, Vivian Do, Yiqun Ma, Tarik Benmarhnia, Arnab K. Dey



its2s framework





its2s

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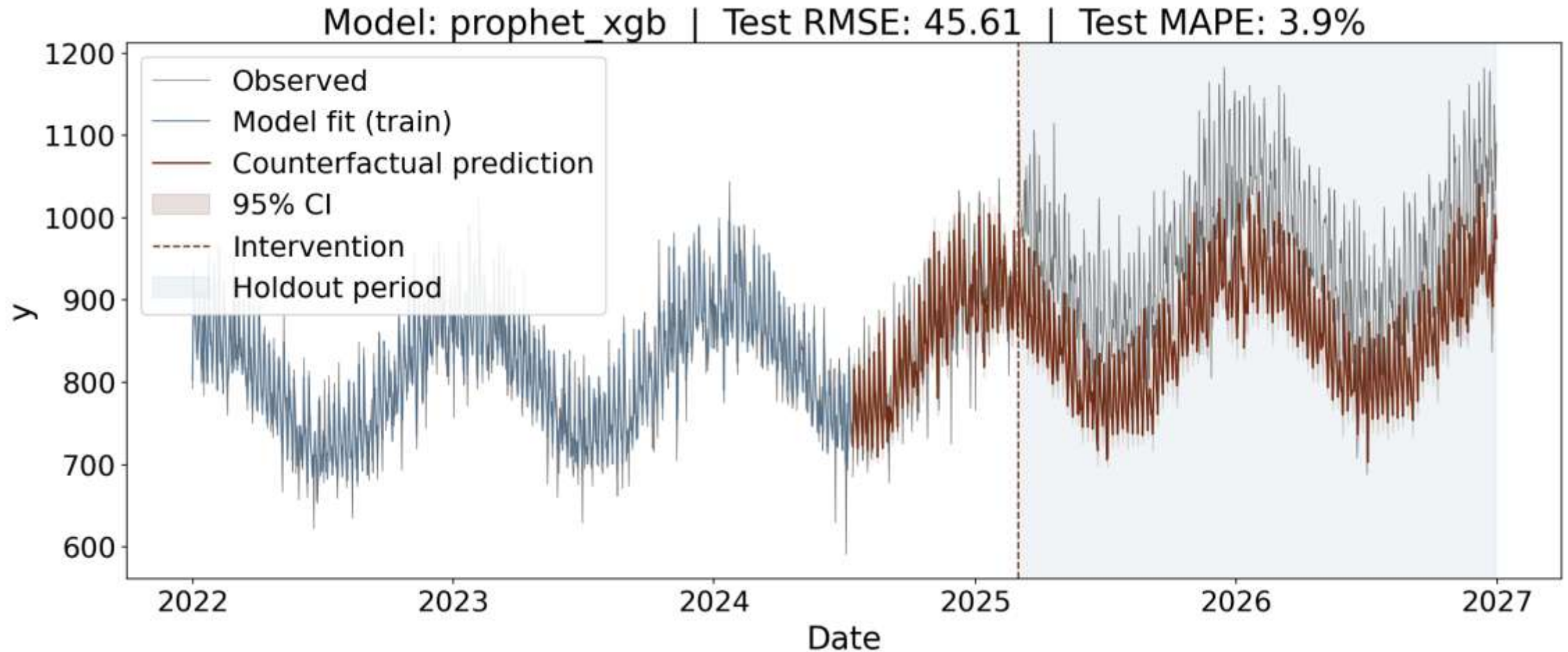
1. An open-source Python package for interrupted time series, 2-stage counterfactual analysis with moving-block bootstrap confidence intervals.
2. A modular framework for fitting ITS models with cross-validation, hyperparameter tuning, and reproducible end-to-end workflows.

Why its2s?

- **Flexibility:** its2s is designed to be flexible, allowing you to easily use this methodology with any data, model, and settings.
- **Modularity:** its2s is designed to be modular, allowing you to use the parts of the pipeline that you need.
- **Reproducibility:** its2s is designed to be reproducible via seeding and rigorous parameter documentation, allowing you to reproduce the results of your analysis easily.
- **Efficiency:** its2s is designed to run quickly and efficiently, allowing you to run the analysis quickly and easily.

[Table of contents](#)[Why its2s?](#)[Core API](#)[Additional functionality](#)[What's next?](#)[Getting help and contributing](#)[Contact](#)

Case study 1: recovering a known effect (+12%)

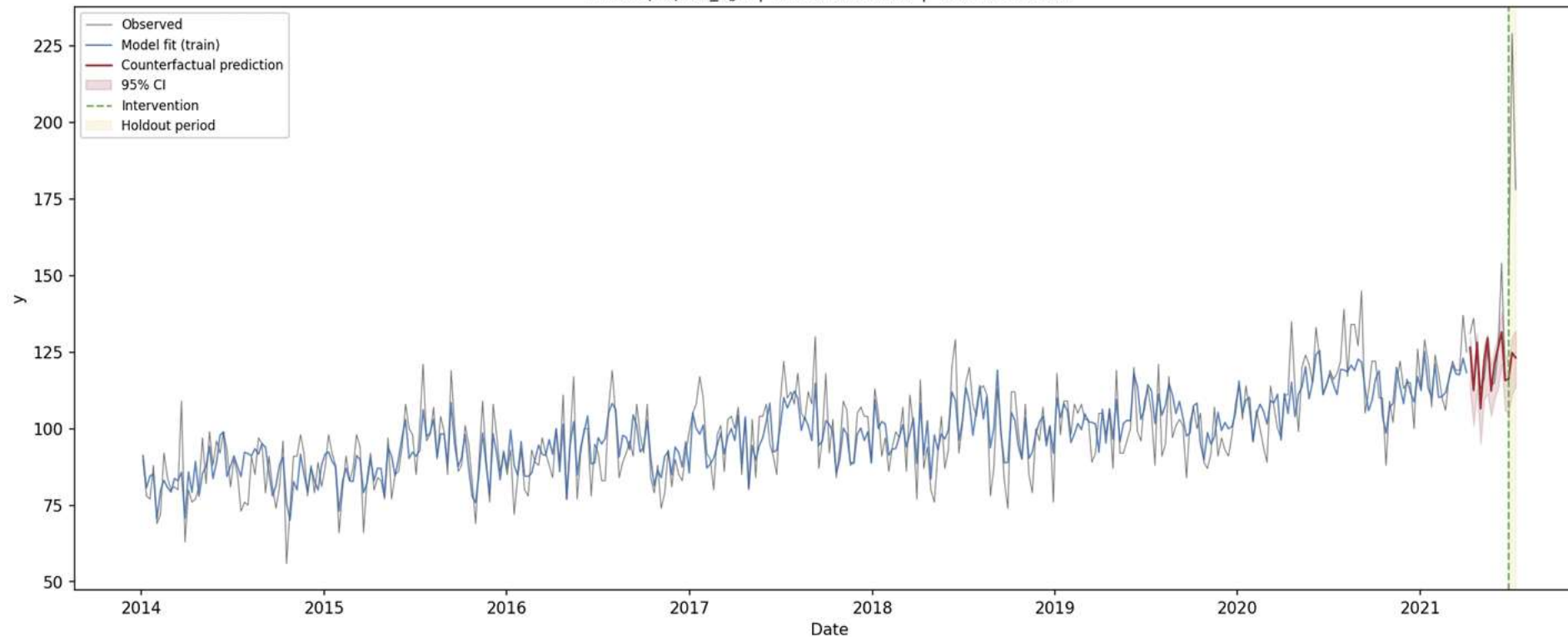


Estimated Excess: + **11.77%**



CASE STUDY 2 – THE 2021 PACIFIC NORTHWEST (PNW) HEAT DOME

Model: prophet_xgb | Test RMSE: 10.57 | Test MAPE: 5.5%

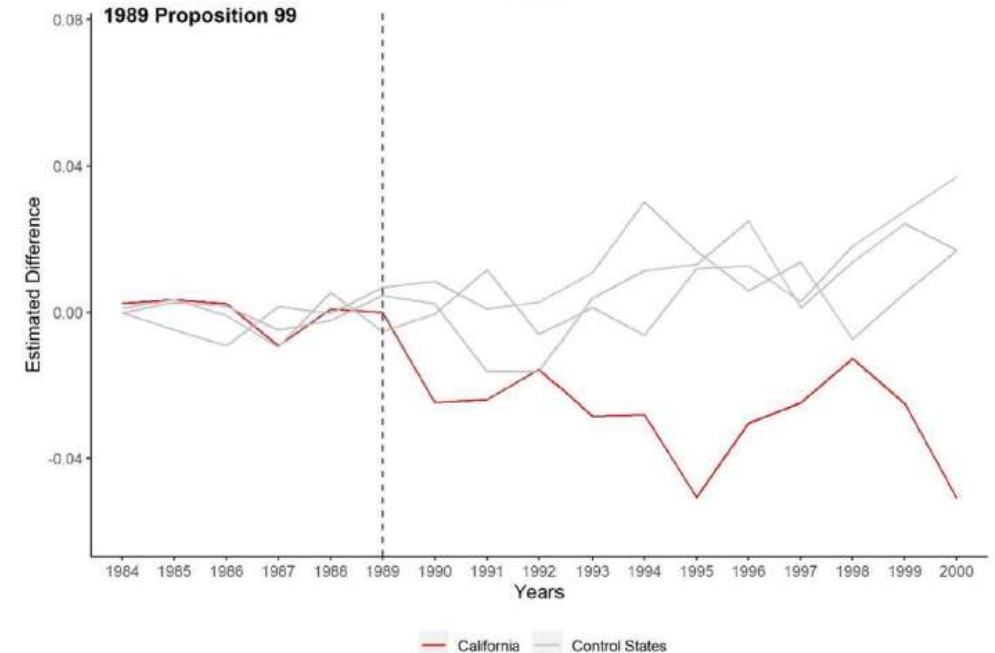


- We estimated 193 excess injury deaths (95% eCI: 179, 217) relative to the counterfactual, corresponding to a 53% excess (eCI: 49%, 60%).



THE IMPORTANCE OF FALSIFICATION TESTS

- Permutation tests
- Many assumptions cannot be checked empirically
 - It is therefore important to design a set of falsification/placebo tests to improve the inference of interest
- Negative control approaches
 - Lipsitch M et al. (2012): Negative controls: a tool for detecting confounding and bias in observational studies.



Sheridan et al. "Evaluating the impact of the California 1995 smoke-free workplace law on population smoking prevalence using a synthetic control method." Preventive medicine reports 19 (2020): 101164.



STAGGERED INTERVENTIONS



MULTIPLE TREATED UNITS AT DIFFERENT TIMES

- A very active area of research
 - Goodman-Bacon (2018) proposed a solution based on a weighted average of all possible DID estimators (using some groups multiple times) in the sample of interest. This approach requires an additional identifying assumption of time-invariant treatment effects.
 - Callaway and Sant'Anna (2020) proposed an analytical solution for such case where there are more than two time periods and units that can become treated at different points in time while relaxing the time-invariant treatment effects assumption



WHAT DO WE MEAN BY STAGGERED INTERVENTIONS?

- When a given policy/treatment is affecting multiple units but at different times points

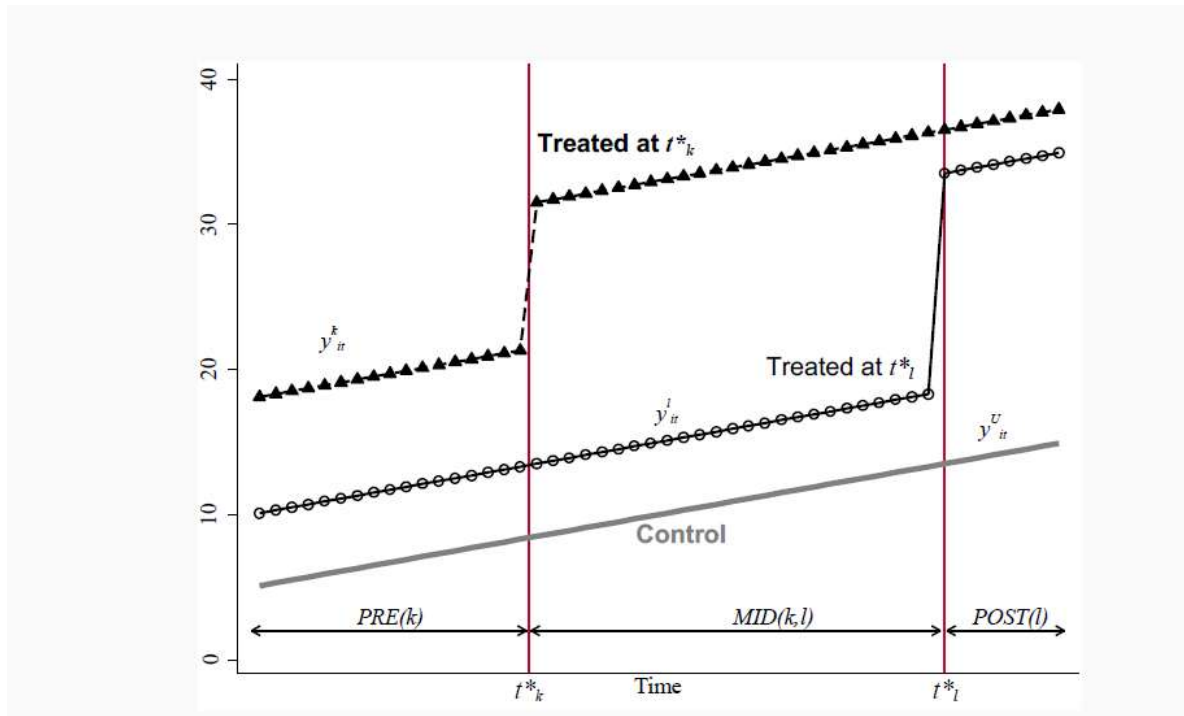


Figure from Goodman-Bacon, 2021



WHY DO TRADITIONAL METHODS FAIL IN SUCH SETTINGS?

- Simple settings for DID and SCM do not apply here for a few reasons:
 - Each treated unit requires a suitable control group specific the timing of the policy/treatment
 - Units that receive the policy/treatment later may be used as control unit earlier in the study period
 - Some potential statistical dependence issues
 - **We do not want to assume that a given policy/treatment has the same effect across units and over time**



ANALYTICAL APPROACHES TO DEAL WITH STAGGERED INTERVENTIONS

- Multiple approaches have been proposed in the past few years
 - A very active area: De Chaisemartin, C., & d'Haultfoeuille, X. (2022)
- We follow the 2-stage approach proposed by Callaway and Sant'Anna (2020)
- This approach requires:
 - To have –enough– never treated units
 - To assume independence regarding the timing of the implementation over treated units
 - Similar assumptions as traditional settings



ITS WITH STAGGERED INTERVENTIONS

Education Corner

Staggered interventions with no control groups

Brice Batomen ^{1,*} and Tarik Benmarhnia^{2,3}

¹Dalla Lana School of Public Health, University of Toronto, Toronto, ON, Canada, ²Scripps Institution of Oceanography, University of California San Diego, La Jolla, CA, USA and ³Irset Institut de Recherche en Santé, Environnement et Travail, Inserm, University of Rennes, EHESP, Rennes, France

*Corresponding author. Department of Epidemiology, Dalla Lana School of Public Health, University of Toronto, 155 College Street, Room 688, Toronto, ON M5T 3M7, Canada. E-mail: brice.kuimi@utoronto.ca

Key Messages

- In staggered intervention scenarios without control groups, common model specification for impact evaluations may yield biased estimates due to ill-defined post-intervention periods.
- Alternative model specifications that are drawn from the difference-in-differences literature for staggered interventions can be easily adopted when no control group is available.
- The adoption of these alternative models improves the validity of impact evaluations, especially if heterogeneity is expected across treated groups and across post-intervention time periods.

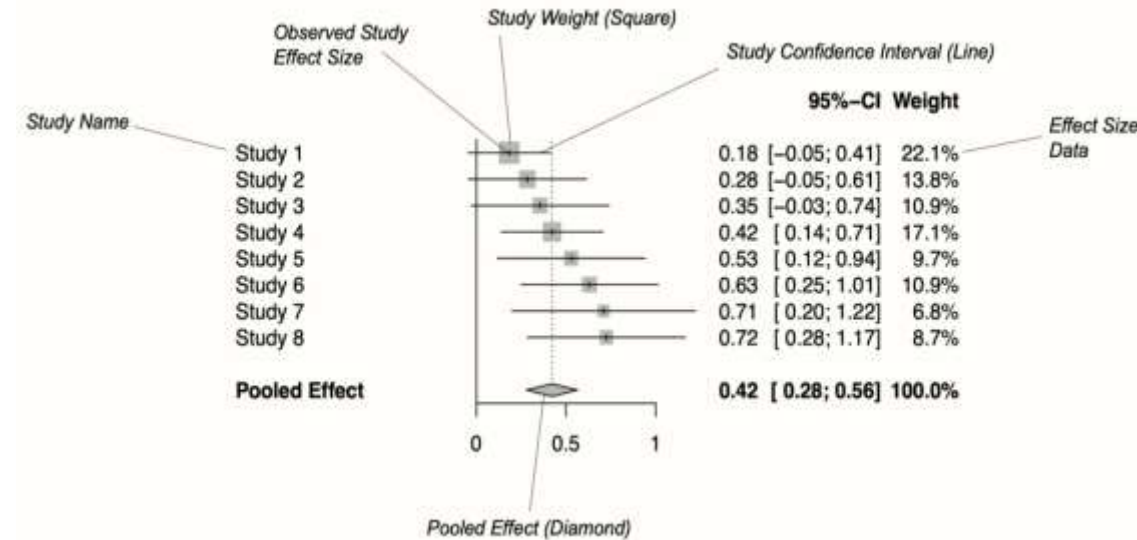
Staggered intervention
with no control groups

$$E(Y) = \alpha + \sum_{g=1}^N \beta_g \text{Group} + \beta_1 \text{Time} + \delta \text{Post}$$
$$E(Y) = \alpha + \sum_{g=1}^N \beta_g \text{Group} + \beta_1 \text{Time} + \sum_{ct} \delta_{ct} \text{Post}_{ct}$$



A QUICK NOTE ON META-ANALYSES AND META-REGRESSIONS

- Meta-analyses are typically used to pool effect estimates from multiple studies
- This is a form of multilevel model, in which observations are nested within studies
- We obtain a pooled effect estimate by applying a weighting procedure (usually based on the inverse of the standard error)
- We can also quantify the level of heterogeneity across studies
- If there is some heterogeneity, we then can conduct a meta-regression in which the dependent variable is the study-specific effect estimate and independent variables study-specific characteristics (e.g. study period, population composition etc...)



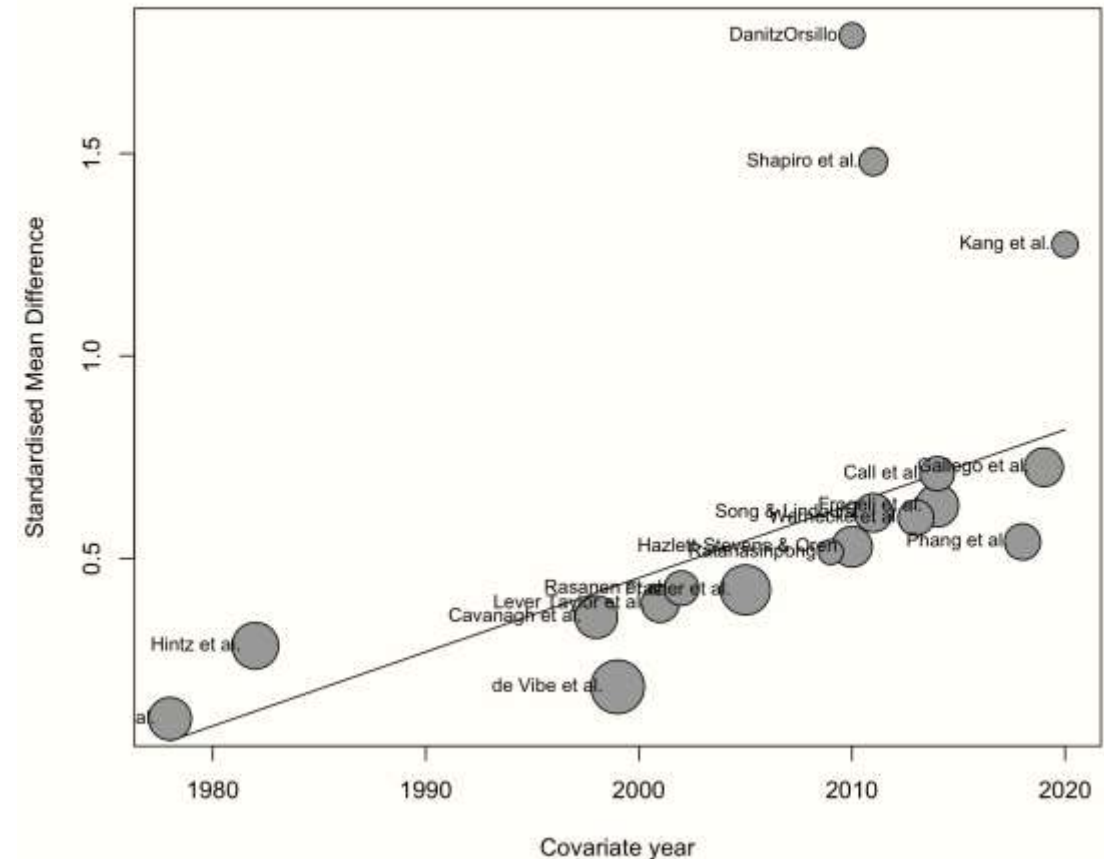
https://bookdown.org/MathiasHarrer/Doing_Meta_Analysis_in_R/forest.html

$$I^2 = 100\% \times \frac{Q - df}{Q}$$



HOW DO META-REGRESSION WORK?

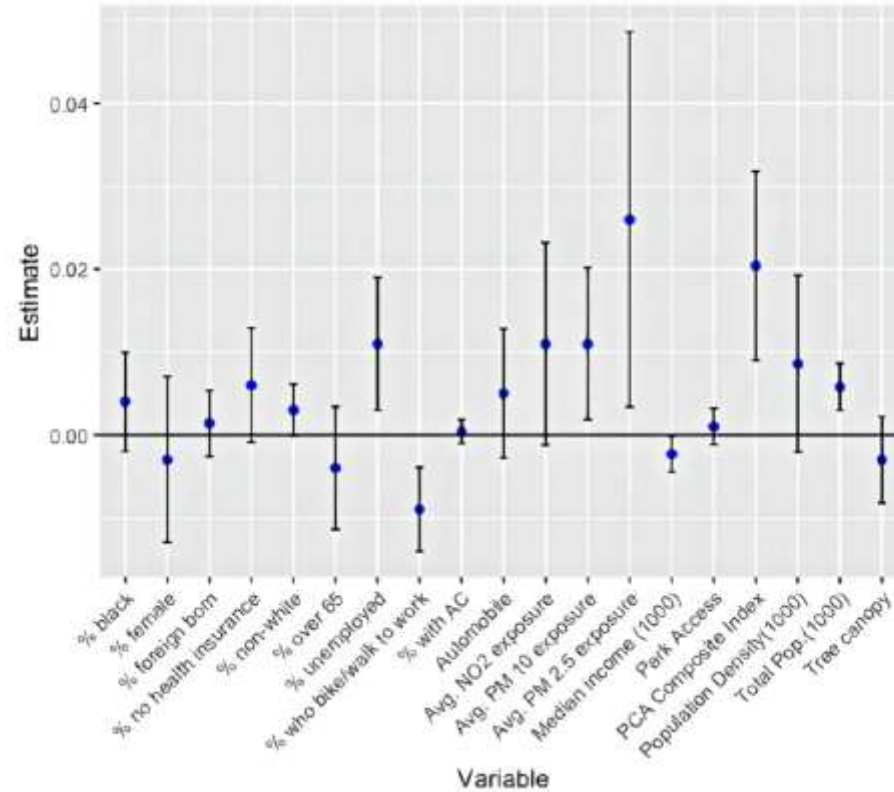
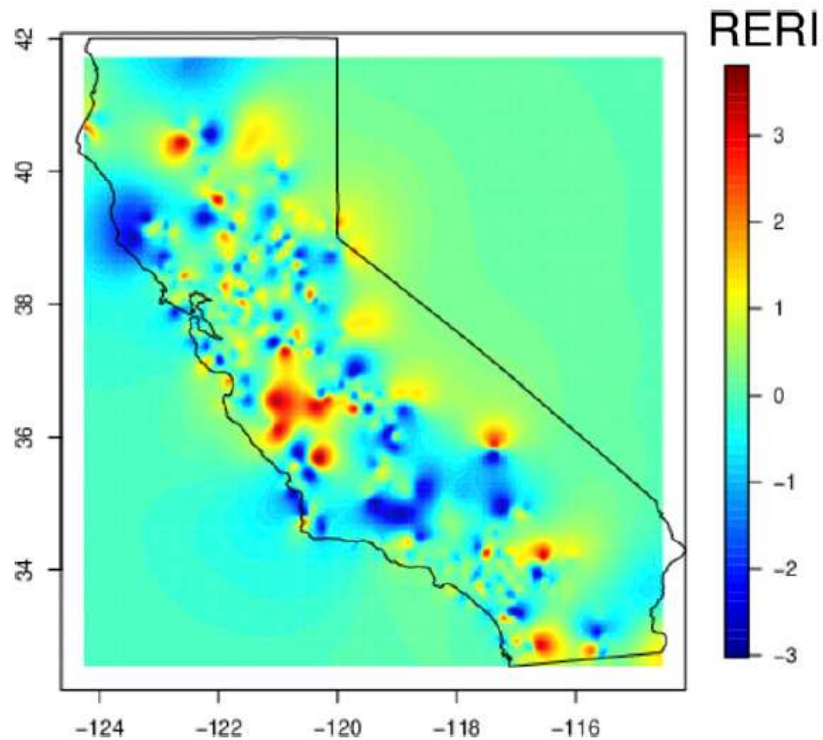
- An intuitive way to describe meta-regression is a weighted regression based on the study-specific variance
 - If all studies have the same variance: meta-regression = simple regression
- We then obtain a slope for each independent variable, and we can understand the drivers of effect estimates heterogeneity



AN ILLUSTRATIVE EXAMPLE

Spatial variation in the joint effect of extreme heat events and ozone on respiratory hospitalizations in California

Lara Schwarz^{a,b,1,2}, Kristen Hansen^{b,1,2}, Anna Alari^c, Sindana D. Ilango^d, Nelson Bernal^a, Rupa Basu^e, Alexander Gershunov^d, and Tarik Benmarhnia^{b,d}

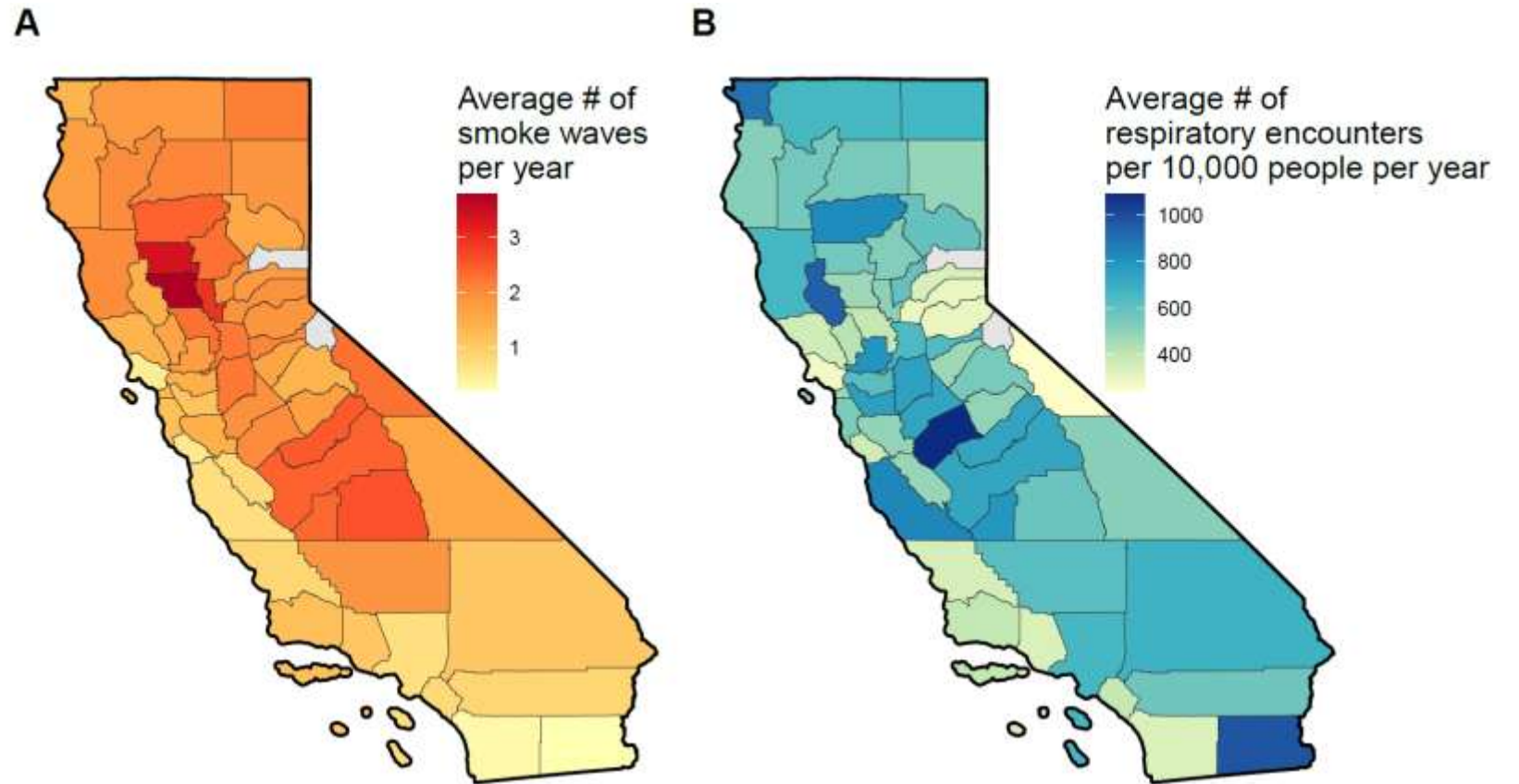


USING 2-STAGE ITS TO UNDERSTAND THE DRIVERS OF MULTIPLE WILDFIRE EVENTS IN CALIFORNIA

(MA ET AL)



In this study, we estimated the excess respiratory acute care utilization during **1,249 smoke wave events** in 56 counties in California, from 2006 to 2019.

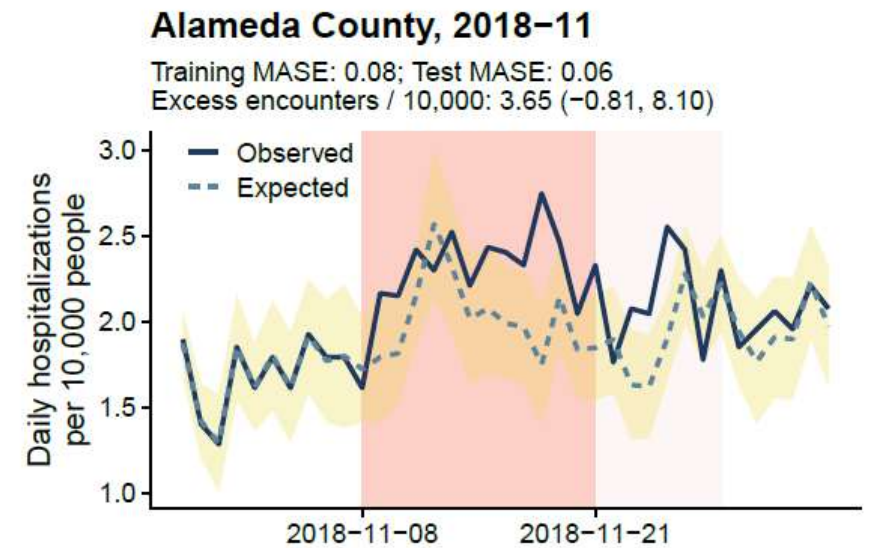
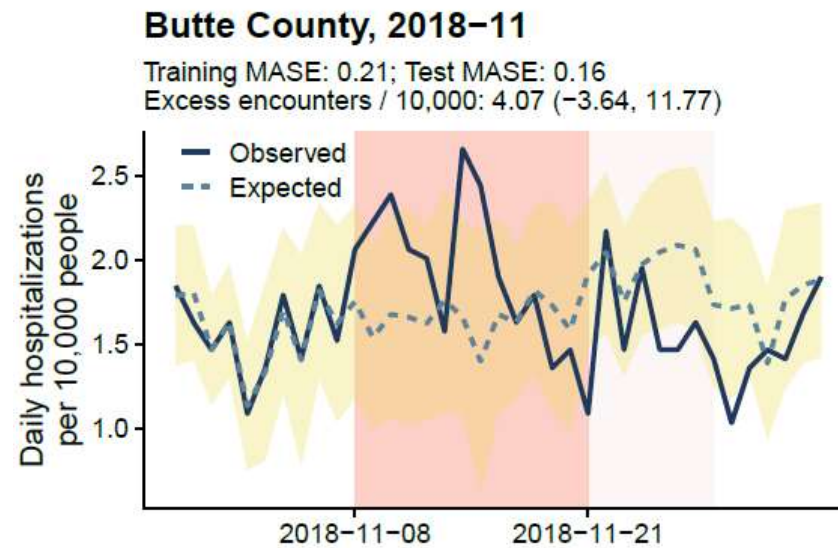
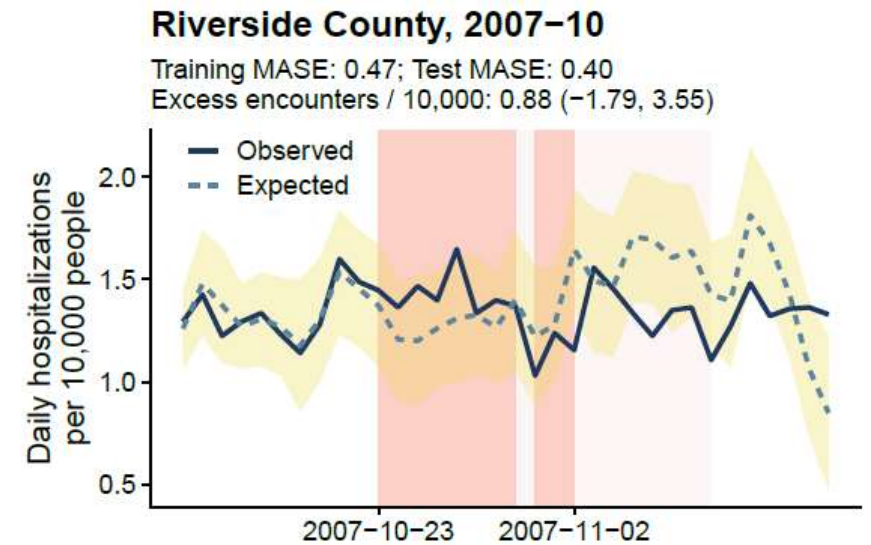
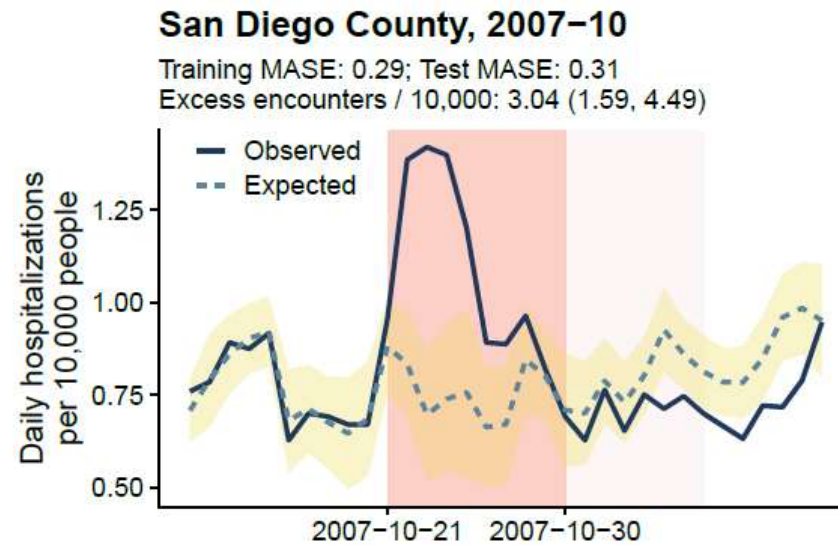


Spatial distribution of county-level smoke waves and respiratory acute care utilization in California, 2006-2019.

Smoke wave: at least 2 consecutive days during which county-level wildfire smoke $PM_{2.5}$ exceeded the 95th percentile of the distribution of daily California-average smoke $PM_{2.5}$ concentrations over the study period ($3.4 \mu g/m^3$)

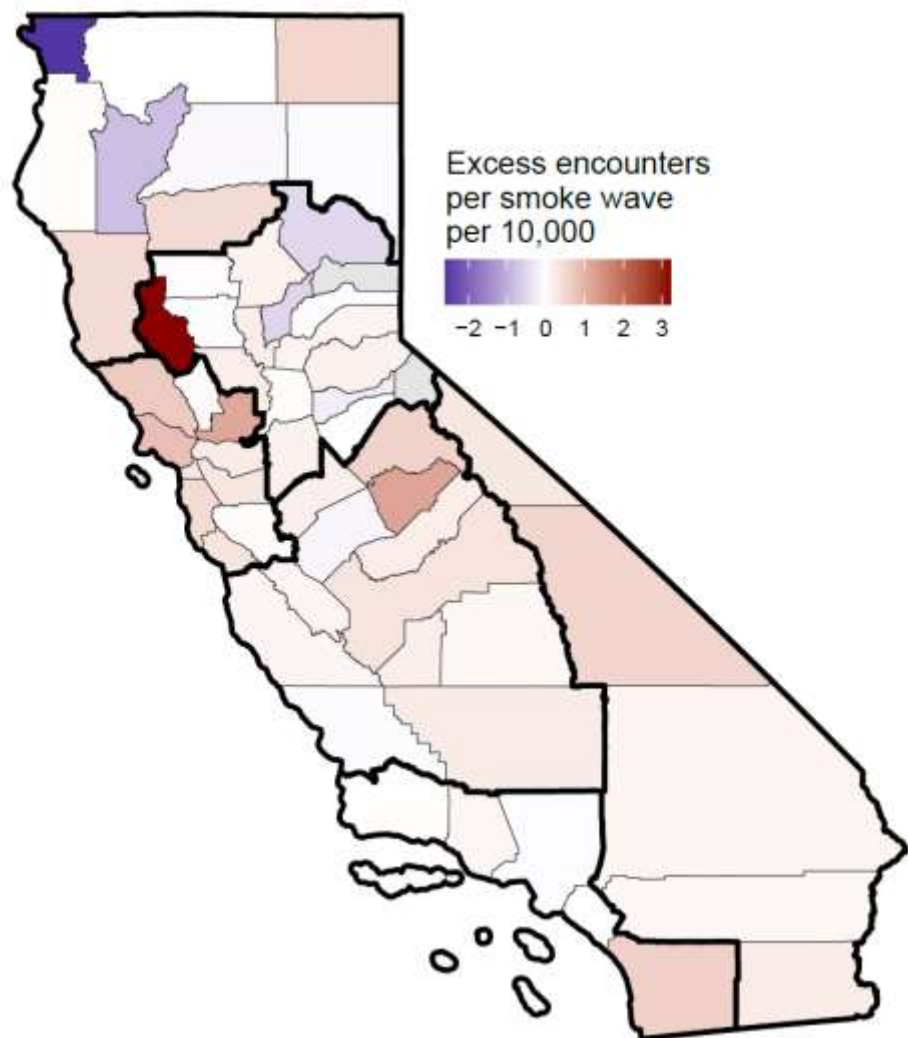
A two-stage interrupted time-series model combined with machine-learning algorithms.

- A Prophet-XGBoost model was used to predict the expected trend of acute care utilization under the counterfactual scenario
- Excess encounters = observed encounters – expected encounters

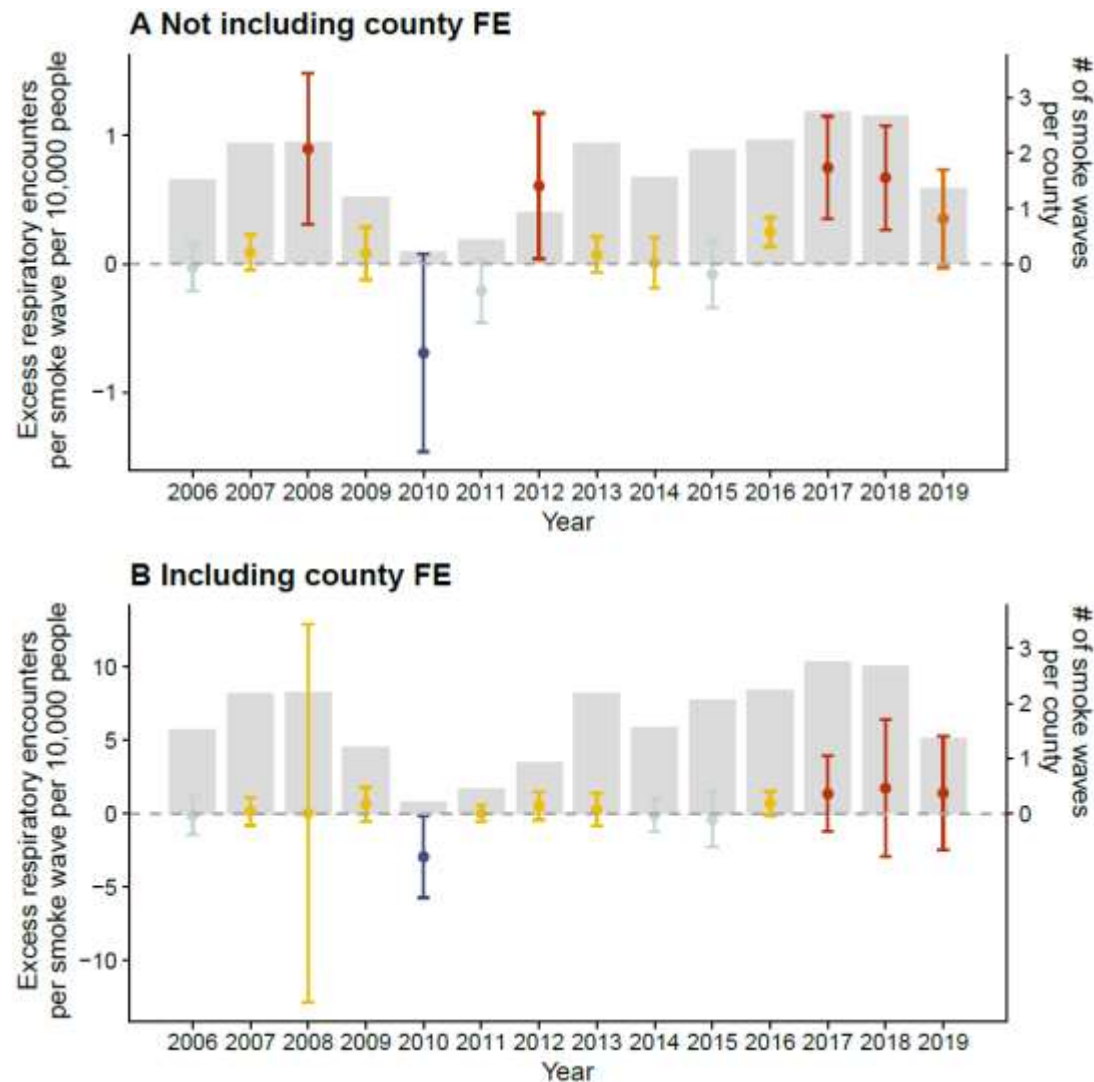


Observed and expected trends of respiratory acute care utilization before, during, and after smoke waves (a few examples)

Important spatial and temporal heterogeneity was observed.

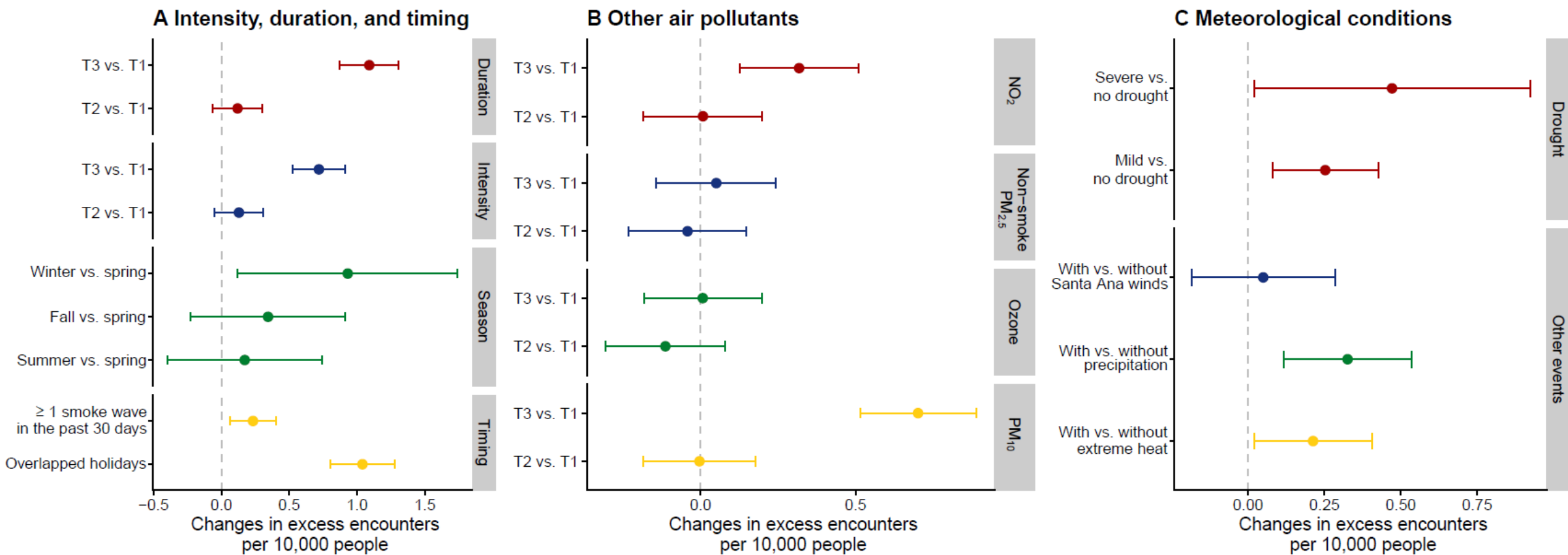


Pooled excess respiratory acute care encounters per smoke wave in each county



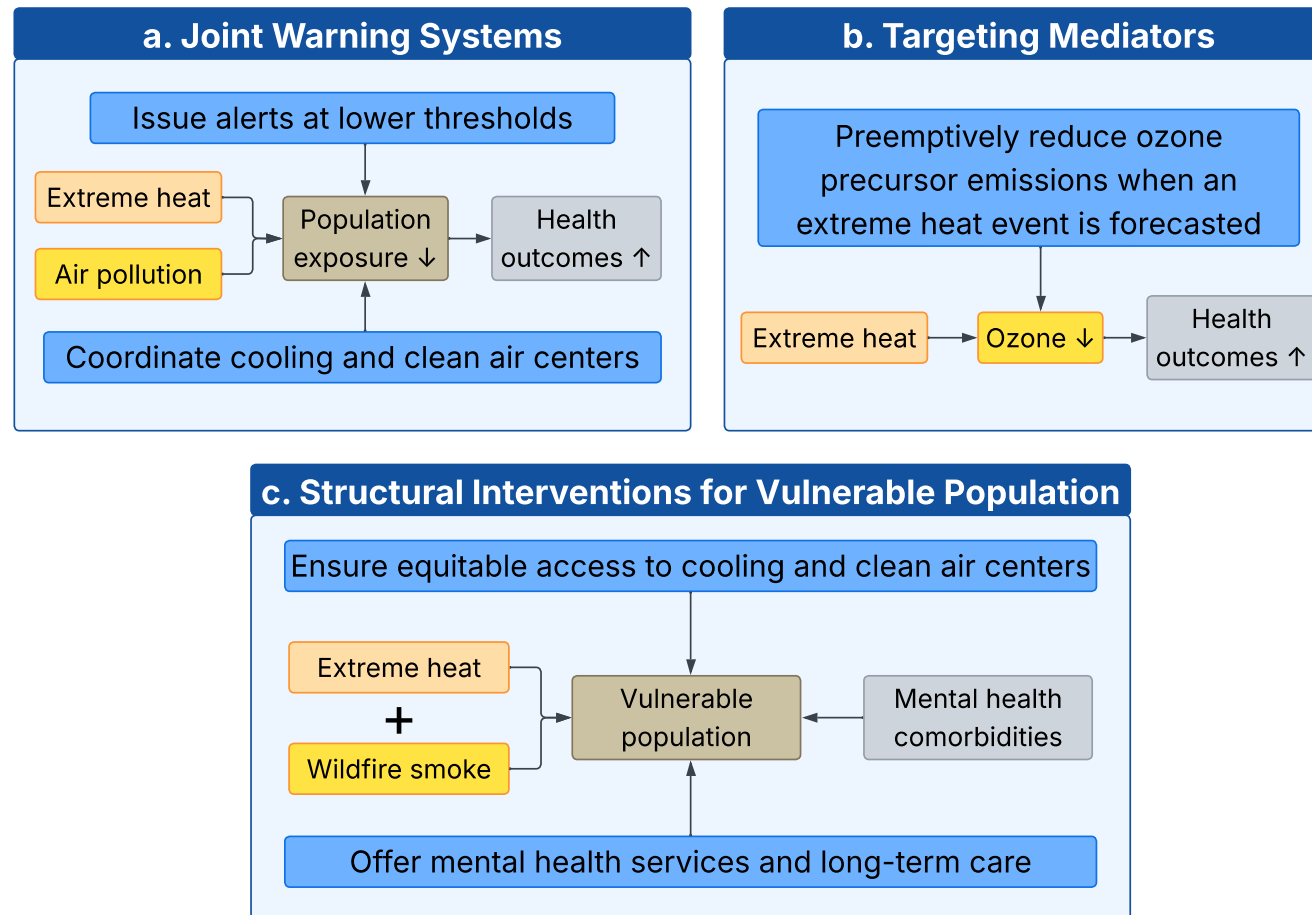
Pooled excess respiratory acute care encounters per smoke wave in each year

The excess acute care utilization were higher when the smoke waves...



Effect modification by smoke wave characteristics and co-occurring environmental conditions

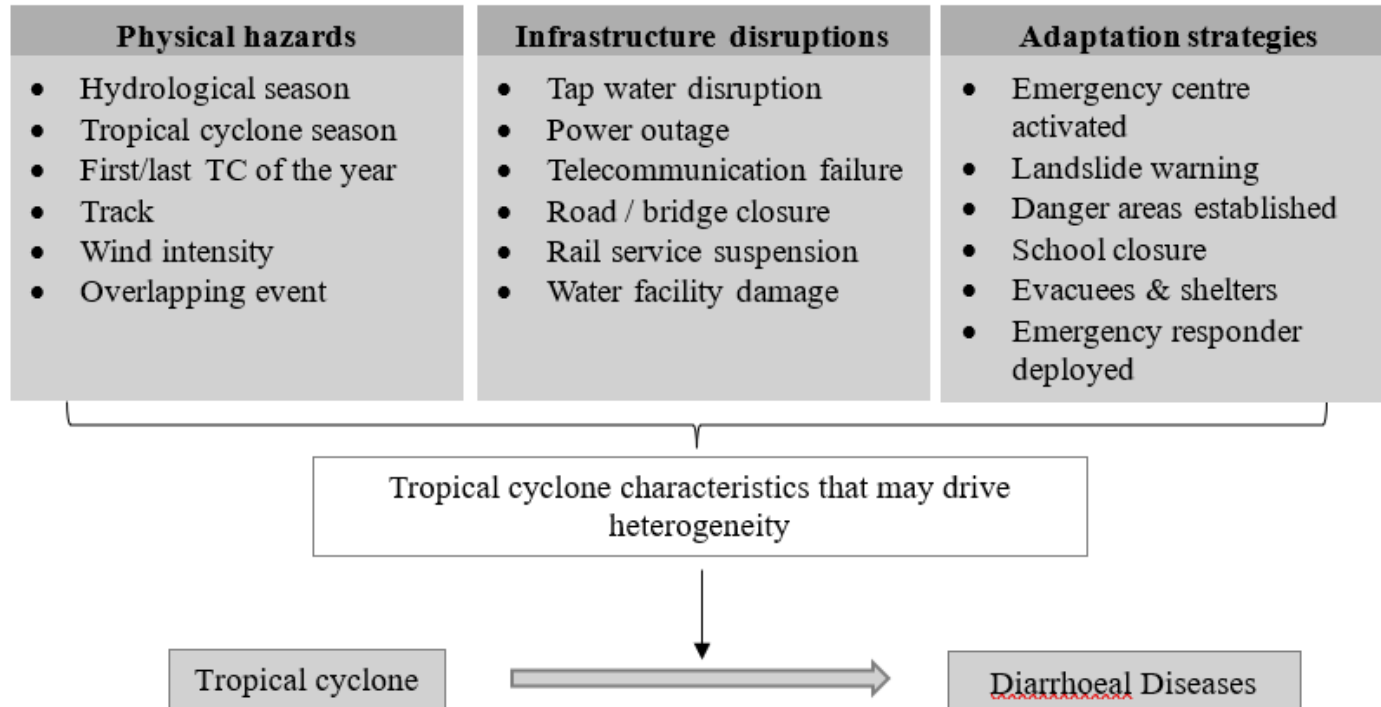
Public health interventions in a compound framework



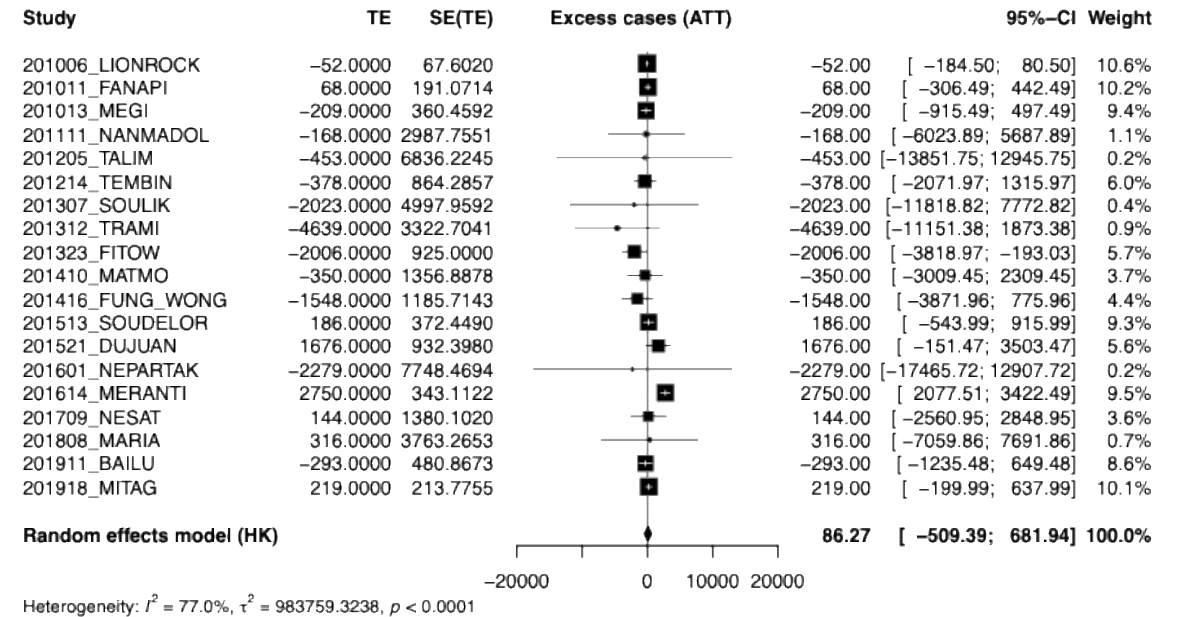
**Quantifying the heterogeneous effects of
tropical cyclones on diarrheal disease in
Taiwan: A two-stage interrupted time series
approach**

Lin et al.

Tropical cyclone characteristics

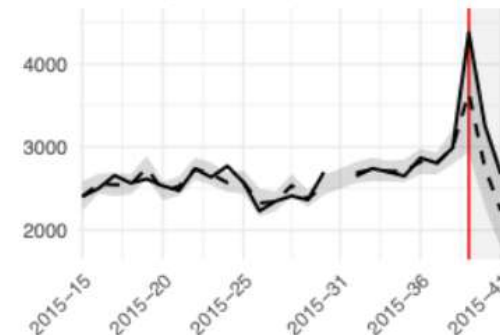


Multiple cyclones events



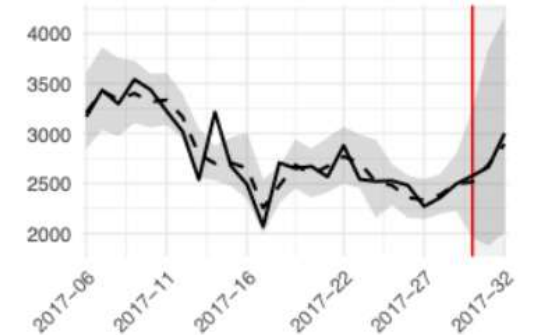
201521_DUJUAN

Excess cases: 1676 (95% CI: -344, 3311)
Rate: 1.5 per 10,000 (95% CI: -0.3, 3.0)

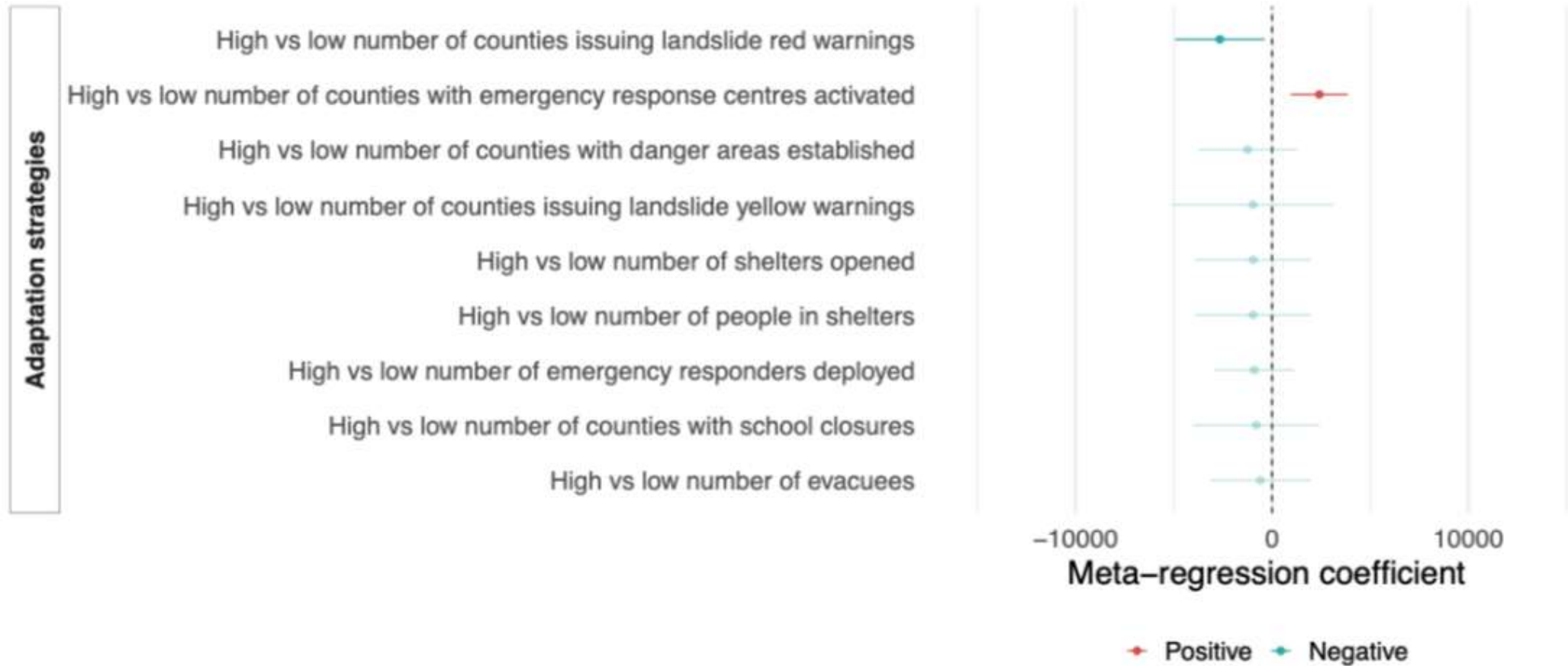


201709_NESAT

Excess cases: 144 (95% CI: -3011, 2399)
Rate: 0.1 per 10,000 (95% CI: -3.0, 2.4)



Drivers of heterogeneous effects of cyclones



Summary

Workshop Part II

1. Introduction to the San Francisco Case Study and Dataset
2. Classical Interrupted Time Series Analysis with Segmented Regression
3. Two-Stage Machine Learning-Integrated Interrupted Time Series Analysis
4. Extension to Multiple Events

THANK YOU

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